CFA Level I Mock Exam 6.2 - Mark Scheme

Each question lists the correct choice and the original feedback for all three choices. Source excerpts accompany explanations where PDF text extraction may lose equation layout.

Question 1

A measure of the sensitivity of a bond's price to a change in a specific maturity segment of the benchmark yield curve best defines:

Correct answer: A

A. key rate duration. - Correct

Correct Answer Feedback: Correct because key rate duration (or partial duration) is a measure of a bond's sensitivity to a change in the benchmark yield curve at a specific maturity segment. In contrast to effective duration, key rate durations help identify 'shaping risk' for a bond—that is, a bond's sensitivity to changes in the shape of the benchmark yield curve.

B. effective duration. - Incorrect

Incorrect Answer Feedback: Incorrect because an approach to assess the interest rate risk of a bond is to estimate the percentage change in price given a change in a benchmark yield curve—for example, the government par curve. This estimate, which is very similar to the formula for approximate modified duration, is called the effective duration. The effective duration of a bond is the sensitivity of the bond's price to a change in a benchmark yield curve. Effective duration measures interest rate risk in terms of a parallel shift in the benchmark yield curve (ΔCurve), not a change in its shape.

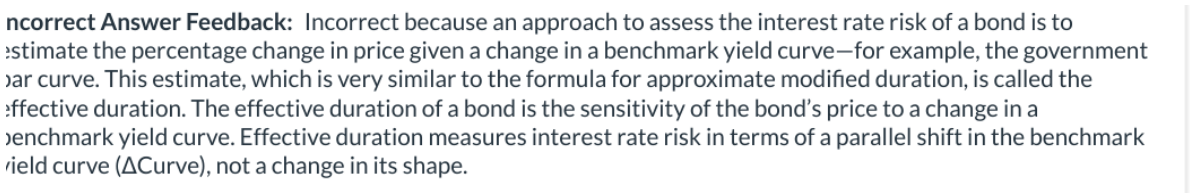

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Figure 1: Source equation for question 1, option B

C. modified duration. - Incorrect

Incorrect Answer Feedback: Incorrect because modified duration has an important application in risk measurement: It provides an estimate of the percentage price change for a bond given a change in its yield-to- maturity.

Question 2

An analyst assumes that a company's market share gains will decline smoothly to zero over the next five years. Which of the following best describes this forecast approach and object?

Correct answer: B

A. Discretionary forecast of a top-down revenue driver - Incorrect

Incorrect Answer Feedback: Incorrect because assuming that a forecast object will converge to zero over the next five years is consistent with the historical base rates and convergence forecasting approach, not the analyst's discretionary forecast approach.

B. Historical base rates and convergence forecast of a top-down revenue driver - Correct

Correct Answer Feedback: Correct because a company's market share is a top-down revenue driver and assuming that this object will converge to zero over the next five years is consistent with the historical base rates and convergence forecasting approach.

C. Historical base rates and convergence forecast of a bottom-up revenue driver - Incorrect

Incorrect Answer Feedback: Incorrect because a company's market share is a top-down, not bottom-up, revenue driver.

Question 3

A portfolio's investment strategy relies on a particular element of the tax code to produce superior after-tax returns for high-net-worth individuals. Because of this strategy, the portfolio most likely faces a high level of:

Correct answer: C

A. legal risk. - Incorrect

Incorrect Answer Feedback: Incorrect because legal risk is the risk of being sued or the risk that a court will not uphold an agreement.

B. model risk. - Incorrect

Incorrect Answer Feedback: Incorrect because model risk is the risk of using the wrong model for analysis or the risk of using the right model incorrectly.

C. compliance risk. - Correct

Correct Answer Feedback: Correct because tax risk, the risk that the tax code could change, along with regulatory and accounting risks together form compliance risk.

Question 4

The expected return for a security is equal to the market's risk premium. If the risk-free rate is positive and the CAPM holds, the beta of the security is:

Correct answer: A

A. less than 1. - Correct

Correct Answer Feedback: Correct because the CAPM equation is E(R) = R + β[E(R) – R]. Therefore, the beta i f m f of the security has to be less than 1 in order for the security's return to be the same as the market risk premium, given that the risk-free rate is positive.

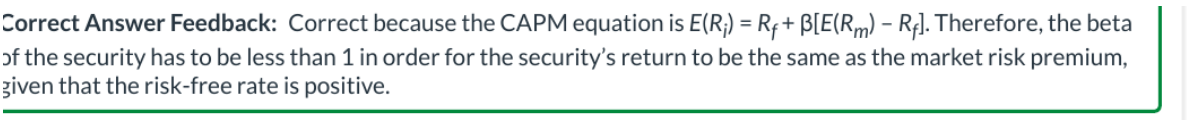


Figure 2: Source equation for question 4, option A

B. equal to 1. - Incorrect

Incorrect Answer Feedback: Incorrect because the CAPM equation is E(R) = R + β[E(R) – R]. Therefore, if beta i f m f is equal to 1, the security's return will be higher than the market risk premium.

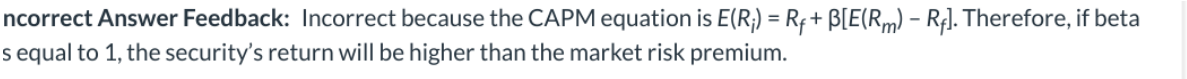


Figure 3: Source equation for question 4, option B

C. greater than 1. - Incorrect

Incorrect Answer Feedback: Incorrect because the CAPM equation is E(R) = R + β[E(R) – R]. Therefore, if beta i f m f is greater than 1, the security's return will be higher than the market risk premium.

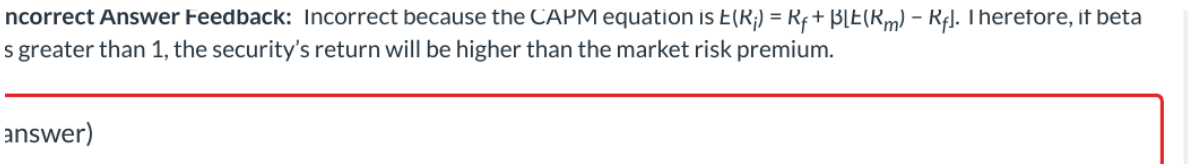


Figure 4: Source equation for question 4, option C

Question 5

With regard to their investment in pension plans, employees with defined contribution plans accept:

Correct answer: C

A. inflation risk only. - Incorrect

Incorrect Answer Feedback: Incorrect because the key to a defined contribution plan is that the employee accepts the investment and inflation risk, not only inflation risk.

B. investment risk only. - Incorrect

Incorrect Answer Feedback: Incorrect because the key to a defined contribution plan is that the employee accepts the investment and inflation risk, not only investment risk.

C. both inflation and investment risks. - Correct

Correct Answer Feedback: Correct because the key to a [defined contribution plan is that the employee accepts the investment and inflation risk and is responsible for ensuring that there are enough assets in the plan to meet their needs upon retirement.

Question 6

Redemption regimes are a feature of:

Correct answer: A

A. covered bonds. - Correct

Correct Answer Feedback: Correct because covered bonds are similar to ABS, but they differ because of their dual recourse nature, strict eligibility criteria, dynamic cover pool, and redemption regimes in the event of sponsor default. Redemption regimes exist to align the covered bond's cash flows as closely as possible with the original maturity schedule in the event of default of a covered bond's financial sponsor.

B. auto loan-backed securities. - Incorrect

Incorrect Answer Feedback: Incorrect because redemption regimes is not a feature of auto loan asset backed securities. In the case of ABS, the financial institution that originates loans transfers securitized assets to a bankruptcy-remote special legal entity, but the pool of underlying assets in a covered bond (or "cover pool") remains on the financial institution's balance sheet, against which covered bondholders retain a top-priority claim. Covered bonds are similar to ABS, but they differ because of their dual recourse nature, strict eligibility criteria, dynamic cover pool, and redemption regimes in the event of sponsor default.

C. commercial mortgage-backed securities. - Incorrect

Incorrect Answer Feedback: Incorrect because redemption regimes is not a feature of commercial mortgage-backed securities. In the case of ABS, the financial institution that originates loans transfers securitized assets to a bankruptcy-remote special legal entity, but the pool of underlying assets in a covered bond (or "cover pool") remains on the financial institution's balance sheet, against which covered bondholders retain a top-priority claim. Covered bonds are similar to ABS, but they differ because of their dual recourse nature, strict eligibility criteria, dynamic cover pool, and redemption regimes in the event of sponsor default.

Question 7

Private equity securities most likely:

Correct answer: A

A. are highly illiquid. - Correct

Correct Answer Feedback: Correct because private equity securities do not have “market determined” quoted prices, are highly illiquid, and require negotiations between investors in order to be traded.

B. trade on active secondary markets. - Incorrect

Incorrect Answer Feedback: Incorrect because private equity securities are not listed on public exchanges, and there is no active secondary market for these securities.

C. have market determined quoted prices. - Incorrect

Incorrect Answer Feedback: Incorrect because private equity securities do not have “market determined” quoted prices.

Question 8

An analyst gathers the following information about a hedge fund:

Beginning-of-year assets under management (AUM)	$100,000,000
Annual gross return	15%
Management fee, based on end-of-year AUM	2%
Incentive fee	20%
Hard hurdle rate	5%

If the incentive fee is based on returns net of management fees, total fees for the year are:

Correct answer: A

A. $3,840,000. - Correct

Correct Answer Feedback: Correct because the partnership agreement usually specifies that the performance (incentive) fee is earned only after the fund achieves a return known as a hurdle rate. The hurdle rate is a minimum rate of return that the GP must exceed in order to earn the performance fee. Ending Value: $100 million × (1 + 15%) = $115 million Management fee based on end of year AUM: $115 million × 2% = $2.30 million Hurdle: $100 million × 5% = $5 million Return after fees and hurdle: $115 million – $100 million – $2.30 million – $5 million = $7.70 million Incentive fee: $7.70 million × 20% = $1.54 million Incentive fee: $7.70 million × 20% = $1.54 million Manager’s total fees: $2.30 million + $1.54 million = $3.84 million.

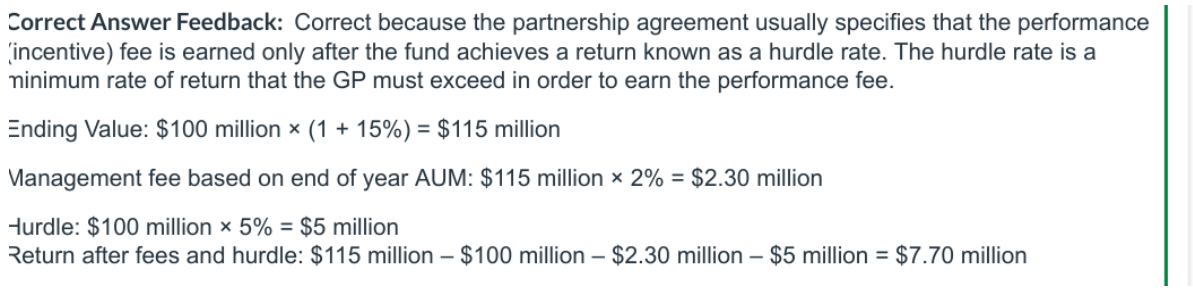


Figure 5: Source equation for question 8, option A

Manager's total fees: $2.30 million + $1.54 million = $3.84 million.

Figure 6: Source equation for question 8, option A

B. $4,300,000. - Incorrect

Incorrect Answer Feedback: Incorrect because it fails to net out management fees from the incentive fee calculation: $100 million × (1 + 15%) = $115 million $115 million × 2% = $2.30 million management fee $100 million × 5% = $5 million hurdle $115 million – $100 million – $5 million = $10 million $10 million × 20% = $2.0 million incentive fee Manager's total fees = $2.30 million + $2.0 million = $4.30 million.

Incorrect Answer Feedback: Incorrect because it fails to net out management fees from the incentive fee calculation:

$100 million × (1 + 15%) = $115 million

$115 million × 2% = $2.30 million management fee

$100 million × 5% = $5 million hurdle
$115 million – $100 million – $5 million = $10 million

$10 million × 20% = $2.0 million incentive fee
Manager's total fees = $2.30 million + $2.0 million = $4.30 million.

Figure 7: Source equation for question 8, option B

C. $4,840,000. - Incorrect

Incorrect Answer Feedback: Incorrect because it omits the hurdle from the incentive fee calculation: $100 million × (1 + 15%) = $115 million $115 million × 2% = $2.30 million management fee $115 million – $100 million – $2.30 million = $12.70 million $12.70 million × 20% = $2.540 million incentive fee Manager's total fees = $2.30 million + $2.54 million = $4.84 million.

ncorrect Answer Feedback: Incorrect because it omits the hurdle from the incentive fee calculation:

$100 million × (1 + 15%) = $115 million

$115 million × 2% = $2.30 million management fee

$115 million – $100 million – $2.30 million = $12.70 million

$12.70 million × 20% = $2.540 million incentive fee
Manager's total fees = $2.30 million + $2.54 million = $4.84 million.

Figure 8: Source equation for question 8, option C

Question 9

The price of a forward contract most likely:

Correct answer: B

A. decreases as the price of the underlying goes up. - Incorrect

Incorrect Answer Feedback: Incorrect. The price of a forward contract is not affected by market conditions. It is set as part of the contract specifications.

B. is constant and set as part of the contract specifications. - Correct

Correct Answer Feedback: Correct. The price of a forward contract remains constant throughout the life of the contract. It is set as part of the contract specifications.

C. increases as market risk increases. - Incorrect

Incorrect Answer Feedback: Incorrect. The price of a forward contract is not affected by market conditions. It is set as part of the contract specifications.

Question 10

With respect to risk governance, which of the following statements is most accurate?

Correct answer: C

A. With more usage of technology, less time is required to test data and models - Incorrect

Incorrect Answer Feedback: Incorrect because with increased reliance on technology, more time and effort must be allotted to test data, models, and results in order to avoid the ironic outcome of the risk of errors coming from within risk systems. within risk systems.

B. Enterprise risk management requires that assets and liabilities be considered in isolation - Incorrect

Incorrect Answer Feedback: Incorrect because enterprise risk management (focusing risk activities on the objectives, health, and value of the whole organization) requires that the entire economic balance sheet of the business be considered, not just the assets or one part of the business in isolation.

C. Both financial assets and human capital can be considered in an individual's risk allocation - Correct

Correct Answer Feedback: Correct because an individual, whose wealth consists not only of financial assets but also of valuable human capital, might prefer that risk allocation decisions be made in view of both forms of capital, optimizing her total wealth.

Question 11

Compared with unregulated markets, regulated markets are best characterized by:

Correct answer: C

A. higher transaction costs. - Incorrect

Incorrect Answer Feedback: Incorrect because regulated markets tend to be more operationally efficient, which leads to lower transactions costs.

B. lower trading volumes. - Incorrect

Incorrect Answer Feedback: Incorrect because regulations help to level the playing field for market participants. When participants believe that markets are fair, they continue to trade in the market, thus increasing trading volumes.

C. reduced arbitrage opportunities. - Correct

Correct Answer Feedback: Correct because regulated markets are more informationally efficient, there are fewer arbitrage opportunities.

Question 12

Challenges in constructing fixed-income indexes most likely include that:

Correct answer: B

A. there is a small number of securities. - Incorrect

Incorrect Answer Feedback: Incorrect because the number of fixed-income securities is many times larger than the number of equity securities. To represent a specific fixed-income market or segment, indexes may include thousands of different securities. Over time, these fixed-income securities mature, and issuers offer new securities to meet their financing needs, leading to turnover in fixed-income indexes.

B. securities are relatively illiquid compared to equities. - Correct

Correct Answer Feedback: Correct because in the fixed income market many securities do not trade frequently and, as a result, are relatively illiquid. As a result, index providers must contact dealers to obtain current prices on constituent securities to update the index or they must estimate the prices of constituent securities using the prices of traded fixed-income securities with similar characteristics.

C. fixed-income markets are predominantly order-driven markets. - Incorrect

Incorrect Answer Feedback: Incorrect because a challenge in index construction is that fixed-income markets are predominantly dealer markets.

Question 13

The first date a buyer of shares is not entitled to an announced dividend is the:

Correct answer: B

A. declaration date. - Incorrect

Incorrect Answer Feedback: Incorrect because first is the declaration date, the day that the company issues a statement declaring a specific dividend. Next comes the ex-dividend date (or exdate), the first date that a share trades without (i.e., “ex”) the dividend.

B. ex-dividend date. - Correct

Correct Answer Feedback: Correct because buyers of a company’s shares on the ex-dividend date are no longer eligible to receive the upcoming dividend, all else being equal, on that day the company’s share price immediately decreases by the amount of the foregone dividend.

C. holder-of-record date. - Incorrect

Incorrect Answer Feedback: Incorrect because buyers of a company’s shares on the ex-dividend date are no longer eligible to receive the upcoming dividend, all else being equal, on that day the company’s share price immediately decreases by the amount of the foregone dividend. This is followed by the holder-of-record date (also called the owner-of-record date, shareholder-of-record date, record date, date of record, or date of book closure), the date that a shareholder listed on the company’s books will be deemed to have ownership of the shares for purposes of receiving the upcoming dividend.

Question 14

Commodity futures prices are most likely in backwardation when:

Correct answer: C

A. interest rates are high. - Incorrect

Incorrect Answer Feedback: Incorrect. Futures price ≈ Spot price (1 + r) + Storage costs – Convenience yield. Thus, high interest rates contribute to an upward sloping commodity forward curve.

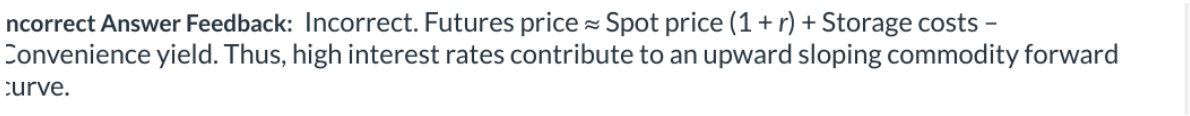


Figure 9: Source equation for question 14, option A

B. storage costs are high. - Incorrect

Incorrect Answer Feedback: Incorrect. Futures price ≈ Spot price (1 + r) + Storage costs – Convenience yield. Thus, high storage costs contribute to an upward sloping commodity forward curve.

ncorrect Answer Feedback: Incorrect. Futures price ≈ Spot price (1 + *r*) + Storage costs – Convenience yield. Thus, high storage costs contribute to an upward sloping commodity forward curve.

Figure 10: Source equation for question 14, option B

C. the convenience yield is high. - Correct

Correct Answer Feedback: Correct. In backwardation, futures prices are lower than spot prices, that is, the commodity forward curve is downward sloping. This scenario occurs when the convenience yield is high. Futures price ≈ Spot price (1 + r) + Storage costs – Convenience yield.

Correct Answer Feedback: Correct. In backwardation, futures prices are lower than spot prices, that is, the commodity forward curve is downward sloping. This scenario occurs when the convenience yield is high. Futures price ≈ Spot price (1 + *r*) + Storage costs – Convenience yield.

answer)

Figure 11: Source equation for question 14, option C

Figure 12: Source equation for question 14, option C

Question 15

In the context of venture capital financing, seed-stage financing most likely supports:

Correct answer: B

A. initial commercial production and sales. - Incorrect

Incorrect Answer Feedback: Incorrect because this activity is supported by early-stage financing, not seed-stage financing. Early-stage financing (early-stage VC), or start-up stage financing, goes to companies moving toward operation but prior to commercial production or sales, in both of which early-stage financing may be injected to initiate.

B. product development and/or marketing efforts. - Correct

Correct Answer Feedback: Correct because seed-stage financing, or seed capital, generally supports product development and marketing efforts, including market research. This is the first stage at which VC funds usually invest.

C. transformation of an idea into a business plan. - Incorrect

Incorrect Answer Feedback: Incorrect because this activity is supported by pre-seed capital, not seed- stage financing. Pre-seed capital, or angel investing, is capital provided at the idea stage. Funds may be used to develop a business plan and to assess market potential.

Question 16

Consider a put option selling for $4 in which the exercise price is $58. What is the profit for a put buyer if the price of the underlying at expiration is $57?

Correct answer: A

A. –$3 - Correct

Correct Answer Feedback: Correct because the put's value at expiration = p = Max(0,X – S), where p is the T T T value of the put at expiration, X is the exercise price, and S is the price of the underlying at expiration. In this T case, Max(0,58 – 57) = $1. The put buyer's profit = Π = p – p (where p is the price of the put at time 0) = 1 – 4 = T 0 0 –$3.

Correct Answer Feedback: Correct because the put's value at expiration = $p_T = Max(0,X - S_T)$, where p_T is the value of the put at expiration, X is the exercise price, and S_T is the price of the underlying at expiration. In this case, $Max(0,58 - 57) = \$1$. The put buyer's profit = $\Pi = p_T - p_0$ (where p_0 is the price of the put at time 0) = 1 – 4 = -$3.

Figure 13: Source equation for question 16, option A

B. $1 - Incorrect

Incorrect Answer Feedback: Incorrect because this is the put's value at expiration. The put's value at expiration = p = Max(S – 0,X), where p is the value of the put at expiration, X is the exercise price, and S is the price of the T T T T underlying at expiration. In this case, Max(0,58 – 57) = $1.

ncorrect Answer Feedback: Incorrect because this is the put's value at expiration. The put's value at expiration = $_T = Max(S_T - 0,X)$, where p_T is the value of the put at expiration, X is the exercise price, and S_T is the price of the underlying at expiration. In this case, $Max(0,58 - 57) = \$1$.

Figure 14: Source equation for question 16, option B

C. $3 - Incorrect

Incorrect Answer Feedback: Incorrect because this is the put seller's profit. The put's value at expiration = –p = T –Max(0,X – S), where p is the value of the put at expiration, X is the exercise price, and S is the price of the T T T underlying at expiration. In this case, –Max(0,58 – 57) = –$1. The put seller's profit = Π = –p + p (where p is the T 0 0 price of the put at time 0) = –1 + 4 = $3.

ncorrect Answer Feedback: Incorrect because this is the put seller's profit. The put's value at expiration = $-p_T = -Max(0,X - S_T)$, where p_T is the value of the put at expiration, X is the exercise price, and S_T is the price of the underlying at expiration. In this case, $-Max(0,58 - 57) = -\$1$. The put seller's profit = $\Pi = -p_T + p_0$ (where p_0 is the price of the put at time 0) = –1 + 4 = $3.

answer)

Figure 15: Source equation for question 16, option C

Question 17

Base-rate neglect is a type of:

Correct answer: C

A. hindsight bias. - Incorrect

Incorrect Answer Feedback: Incorrect because base-rate neglect is a representativeness bias, not a hindsight bias. Hindsight bias refers to believing past events as having been predictable and reasonable to expect.

B. endowment bias. - Incorrect

Incorrect Answer Feedback: Incorrect because base-rate neglect is a representativeness bias, not an endowment bias. Endowment bias is an emotional bias in which people value an asset more when they own it than when they do not.

C. representativeness bias. - Correct

Correct Answer Feedback: Correct because base-rate neglect and sample-size neglect are two types of representativeness bias that apply to FMPs (financial market participants). In base-rate neglect, a phenomenon's rate of incidence in a larger population—its base rate—is neglected in favor of specific information.

Question 18

A mechanism that allows a hedge fund to temporarily restrict redemptions by all investors is known as a:

Correct answer: A

A. gate. - Correct

Correct Answer Feedback: Correct because in addition, funds sometimes impose a gate, which limits or restricts redemptions for a period of time. A gate is imposed by the hedge fund manager independent of any lockup period that may apply. A gate therefore applies to all investors for a finite period of time while a lockup period applies to new investors only.

B. notice period. - Incorrect

Incorrect Answer Feedback: Incorrect because notice periods provide an opportunity for the hedge fund manager to liquidate a position in an orderly fashion without magnifying the losses. A notice period applies to all investors wanting to withdraw some or all of their capital from a hedge fund with the goal of providing the manager with advance warning of the withdrawal.

C. lockup period. - Incorrect

Incorrect Answer Feedback: Incorrect because lockup periods—time periods when investors cannot withdraw their capital—provide the hedge fund manager the required time to implement and potentially realize a strategy's expected results. Lockup periods apply to new investors in a hedge fund with the goal of allowing the hedge fund manager time to implement the fund's investment strategy.

Question 19

The partnership agreement of a private equity fund has no catch-up clause and includes the following:

Hurdle rate	9.0%
Performance fee	20.0%

Profits generated by the fund are allocated between limited partners and the general partner according to an 80/20 split. If the general partner has achieved an IRR of 22.0% for an investment, the general partner's return from the performance fee is closest to:

Correct answer: B

A. 1.8%. - Incorrect

Incorrect Answer Feedback: Incorrect because this answer choice is simply the product of the hurdle rate and the performance fee or 0.09 x 0.20 = 0.018 = 1.8%.

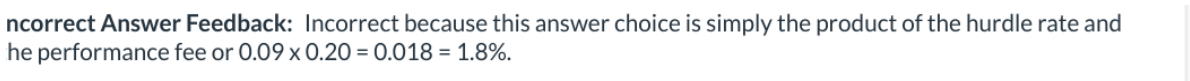


Figure 16: Source equation for question 19, option A

B. 2.6%. - Correct

Correct Answer Feedback: Correct because without a catch-up clause, the LPs will receive the entirety of the first 9% profit (up to the hurdle rate). The remaining 13% will be split 80/20 between the LPs and GP, reducing the GP's return to 2,6%, based on the following formula: (IRR - hurdle rate) x performance fee = (22% – 9%) x 20% = 2.6%.

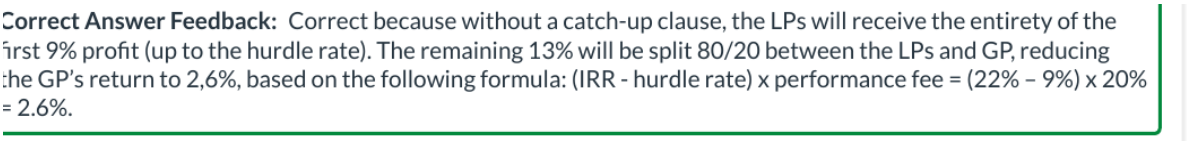


Figure 17: Source equation for question 19, option B

C. 4.4%. - Incorrect

Incorrect Answer Feedback: Incorrect because this is the profit earned by the GP with, not without, a catch up clause. With a catch up clause, the LPs earn the first 9% (up to the hurdle rate) and the GP earns their 20% performance fee on the remaining 13% before any split = 13% x 20% = 2.2%. Remaining profits are then split 80/20. The GP receives (22% – 9% – 2.2%) x 20% = 2.16%. The GP's return = 2.2% + 2.16% = 4.36% ≈ 4.4%.

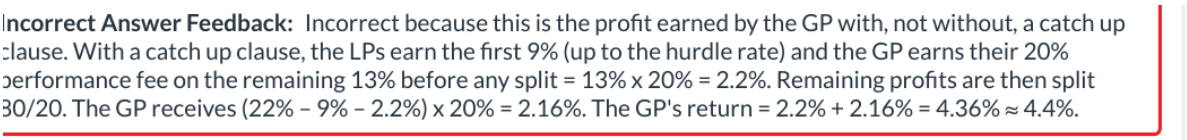


Figure 18: Source equation for question 19, option C

Question 20

If a floating-rate note is trading at a discount to par, this bond's required margin is:

Correct answer: C

A. less than the quoted margin. - Incorrect

Incorrect Answer Feedback: Incorrect because if the required margin is less than the quoted margin, the note will trade at a premium to par.

B. the same as the quoted margin. - Incorrect

Incorrect Answer Feedback: Incorrect because if the required margin is the same as the quoted margin, the note would trade at par value.

C. greater than the quoted margin. - Correct

Correct Answer Feedback: Correct because if the required margin is greater than the quoted margin, the note will trade at a discount to par. For example on a floater with a margin of 50 bps, if the required margin goes up to 75 bps because of a downgrade in the issuer's credit rating, a floater having a quoted margin of 50 bps now pays its investors a "deficient" interest payment. This FRN will be priced at a discount below par value.

Question 21

A characteristic of real assets is that they most likely:

Correct answer: C

A. trade in liquid markets. - Incorrect

Incorrect Answer Feedback: Incorrect because real assets tend to trade in very illiquid markets. B. are inexpensive to manage.

B. are inexpensive to manage. - Incorrect

Incorrect Answer Feedback: Incorrect because investments in real assets generally require substantial management to ensure that the assets are maintained and used efficiently. Investment managers investing in such assets must either hire personnel to manage them or hire outside management companies. Either way, management of real assets is quite costly.

C. are unique assets with different attributes. - Correct

Correct Answer Feedback: Correct because real assets are unique properties in the sense that no two properties are alike.

Question 22

A repo is best defined as a:

Correct answer: C

A. publicly traded, collateralized short-term security. - Incorrect

Incorrect Answer Feedback: Incorrect because a repurchase agreement or repo is the sale of a security with a simultaneous agreement by the seller to buy the same security back from the purchaser at an agreed-on price and future date. It is a private transaction between parties and not publicly-traded.

B. security that provides the holder the right to sell it back to the issuer at par. - Incorrect

Incorrect Answer Feedback: Incorrect because this describes a putable bond, not a repo transaction. A putable bond allows the investor to sell the bond back to the issuer prior to maturity, usually at par value.

C. sale of a security with an agreement by the seller to buy it back at a specified price. - Correct

Correct Answer Feedback: Correct because a repurchase agreement or repo is the sale of a security with a simultaneous agreement by the seller to buy the same security back from the purchaser at an agreed-on price and future date.

Question 23

Compared to mutual funds, hedge funds most likely:

Correct answer: B

A. have increased operating constraints. - Incorrect

Incorrect Answer Feedback: Incorrect because with reduced operating constraints, hedge funds may invest in less liquid and unnoticed opportunities, the true valuation of which may at times be opaque.

B. have a longer notice period for redemptions than a mutual fund. - Correct

Correct Answer Feedback: Correct because investors in modern hedge funds are subject to extended holding periods (known as lockup periods) and subsequent notice periods before an investment redemption is possible. Compared, for example, to many mutual funds allowing easier redemption and guaranteed liquidity on one day's notice, such mandatory periods combine with a lighter regulatory compliance burden (in certain areas) to allow hedge funds more flexibility.

C. use a traditional index benchmark to gauge investment performance. - Incorrect

Incorrect Answer Feedback: Incorrect because a contemporary hedge fund generally has a goal to generate high returns, either in an absolute sense or on a risk-adjusted basis relative to its portfolio-level volatility. Gauging performance relative to a traditional index benchmark can be difficult, given that low correlation with traditional asset investing is frequently a selling point.

Question 24

A company announces an unexpected improvement in its earnings forecast for the coming year. The announcement most likely immediately impacts:

Correct answer: B

A. the company's book value of equity only. - Incorrect

Incorrect Answer Feedback: Incorrect because book value reflects the difference between a company's total assets and total liabilities and not expectations regarding future earnings. The book value of a company's equity reflects the historical operating and financing decisions of its management.

B. the company's market value of equity only. - Correct

Correct Answer Feedback: Correct because market value reflects the collective and differing expectations of investors concerning the amount, timing, and uncertainty of the company's future cash flows, and an unexpected change in a company's earnings forecast will change investors expectations.

C. both the company's book value of equity and the company's market value of equity. - Incorrect

Incorrect Answer Feedback: Incorrect because book value reflects the difference between the company's total assets and total liabilities and not expectations regarding future earnings. The book value of a company's equity reflects the historical operating and financing decisions of its management. The market value of the company's equity reflects these decisions as well as investors' collective assessment and expectations about the company's future cash flows generated by its positive net present value investment opportunities.

Question 25

Which of the following sections of an IPS most likely provides information about specific types of assets to be excluded from a portfolio?

Correct answer: B

A. Procedures - Incorrect

Incorrect Answer Feedback: Incorrect because the Procedures section explains the steps to take to keep the IPS current and the procedures to follow to respond to various contingencies.

B. Investment Guidelines - Correct

Correct Answer Feedback: Correct because the Investment Guidelines section provides information about how policy should be executed (e.g., on the permissible use of leverage and derivatives) and on specific types of assets excluded from investment, if any.

C. Statement of Duties and Responsibilities - Incorrect

Incorrect Answer Feedback: Incorrect because the Statement of Duties and Responsibilities section details the duties and responsibilities of the client, the custodian of the client's assets, and the investment managers.

Question 26

Only the seller can default on a(n):

Correct answer: B

A. swap. - Incorrect

Incorrect Answer Feedback: Incorrect because in contrast to forwards and swaps, in which either party could default to the other, default in options is possible only from the short to the long. Since a swap is equivalent to a series of forward contracts, default in swaps can occur from the long to the short and from the short to the long.

B. option. - Correct

Correct Answer Feedback: Correct because the option buyer (the long) does not have to exercise the option, beyond the initial payment of the premium, therefore there is no obligation of the long to the short. Thus, only the short can default, which would occur if the long exercises the option and the short fails to do what it is supposed to do. Thus, in contrast to forwards and swaps, in which either party could default to the other, default in options is possible only from the short to the long.

C. forward. - Incorrect

Incorrect Answer Feedback: Incorrect because in contrast to forwards and swaps, in which either party could default to the other, default in options is possible only from the short to the long. Therefore, default in forwards can occur from the long to the short and from the short to the long.

Question 27

Which of the following statements regarding alternative asset co-investing is correct? • Statement 1: The investor invests in assets indirectly through a fund. • Statement 2: The investor possesses rights to invest directly in the same assets

Correct answer: C

A. Statement 1 only - Incorrect

Incorrect Answer Feedback: Incorrect because both statements are correct. In co-investing, an investor invests in assets indirectly through the fund and has rights to invest in the same assets directly without an intermediary. Many investors gain direct investing experience via co-investing, where the investor invests in assets indirectly through the fund but also possesses rights (known as co-investment rights) to invest directly in the same assets. Direct investing occurs when an investor makes a direct investment in an asset without the use of an intermediary.

B. Statement 2 only - Incorrect

Incorrect Answer Feedback: Incorrect because both statements are correct. In co-investing, an investor invests in assets indirectly through the fund and has rights to invest in the same assets directly without an intermediary. Many investors gain direct investing experience via co-investing, where the investor invests in assets indirectly through the fund but also possesses rights (known as co-investment rights) to invest directly in the same assets. Direct investing occurs when an investor makes a direct investment in an asset without the use of an intermediary.

C. Both Statement 1 and Statement 2 - Correct

Correct Answer Feedback: Correct because both statements are correct. In co-investing, an investor invests in assets indirectly through the fund and has rights to invest in the same assets directly without an intermediary. Many investors gain direct investing experience via co-investing, where the investor invests in assets indirectly through the fund but also possesses rights (known as co-investment rights) to invest directly in the same assets. Direct investing occurs when an investor makes a direct investment in an asset without the use of an intermediary.

Question 28

An analyst gathers the following information (in $ millions) about two companies for the year just ended:

	Company X	Company Y
Operating income	10	12
Depreciation and amortization	2	1
Interest expense	1	3
Debt	50	60

Based on an assessment of leverage, Company X has:

Correct answer: C

A. lower creditworthiness than Company Y. - Incorrect

Incorrect Answer Feedback: Incorrect because Company X has higher, not lower, creditworthiness than Company Y. The candidate in this case subtracts instead of adding back depreciation and amortization to calculate EBITDA, and calculates EBITDA for Company X as EBITDA = 10 – 2 = 8 and its leverage ratio as Debt/EBITDA = 50/8 = 6.25. For Company Y, the candidate calculates EBITDA as EBITDA = 12 – 1 = 11 and its leverage ratio as Debt/EBITDA = 60/11 ≈ 5.4545 ≈ 5.45, indicating incorrectly that Company X has a higher leverage ratio and thus lower creditworthiness than Company Y.

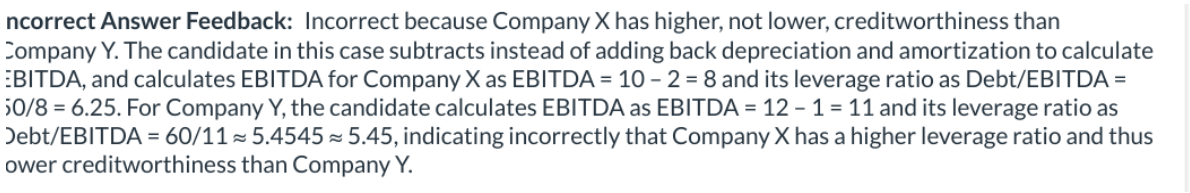


Figure 19: Source equation for question 28, option A

B. the same creditworthiness as Company Y. - Incorrect

Incorrect Answer Feedback: Incorrect because Company X has higher creditworthiness than Company Y. The candidate in this case omits to add back depreciation and amortization to calculate EBITDA, and calculates the leverage ratio for Company X as Debt/EBIT = 50/10 = 5 and the leverage ratio for Company Y as Debt/EBIT = 60/12 = 5, indicating incorrectly that the two companies have the same creditworthiness.

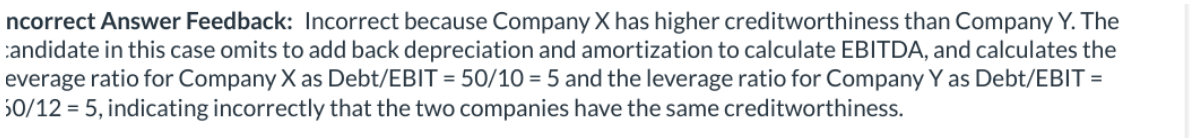


Figure 20: Source equation for question 28, option B

C. higher creditworthiness than Company Y. - Correct

Correct Answer Feedback: Correct because Company X's EBITDA is calculated as EBITDA = operating income + depreciation + amortization = 10 + 2 = 12 and Debt/EBITDA = 50/12 ≈ 4.1667 ≈ 4.2. Company Y's EBITDA is calculated as EBITDA = 12 + 1 = 13 and Debt/EBITDA = 60/13 ≈ 4.6154 ≈ 4.6. Company X has a lower leverage ratio, and thus higher creditworthiness than Company Y.

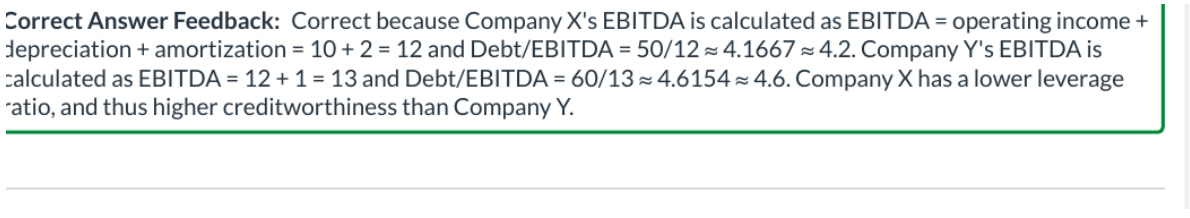


Figure 21: Source equation for question 28, option C

Question 29

An analyst gathers the following information about a company for its fiscal year:

Net profit margin	8%
Total asset turnover	1.5
Financial leverage	1.2

Net profit margin	8%
EPS	€0.15
Dividend per common share	€0.10

The sustainable growth rate is closest to:

Correct answer: B

A. 3.2%. - Incorrect

Incorrect Answer Feedback: Incorrect because it assumes that ROE = net profit margin × leverage (omitting asset turnover); or = 8% × 1.2 = 0.096. Substituting into the sustainable growth rate = retention rate × ROE = (1 – 0.10/0.15) × 0.096 = 0.0320 = 3.2%.

B. 4.8%. - Correct

Correct Answer Feedback: Correct because the sustainable growth rate = retention rate × ROE, where retention rate = (1 – payout ratio), and the payout ratio is the percentage of earnings that the company pays out as dividends to shareholders, or (EPS – DPS) / EPS, and using the DuPont analysis ROE = net profit margin × total asset turnover × leverage. The retention rate = 1 – (0.10 / 0.15) = 1 – 0.667 = 0.3333. ROE = 8% × 1.5 × 1.2 = 0.144. Substituting into the sustainable growth rate formula: 0.3333 × 0.144 = 0.04799 ≈ 4.8%.

C. 9.6%. - Incorrect

Incorrect Answer Feedback: Incorrect because it uses the payout ratio, not retention rate, when calculating the sustainable growth rate (sustainable growth rate = retention rate × ROE). Based on DuPont analysis (ROE = net profit margin × total asset turnover × leverage), ROE = 8% × 1.2 × 1.5, or 0.144; multiplying by the payout ratio = 0.144 × (0.10 / 0.15) = 0.096 = 9.6%.

Question 30

An increase in the dividend payout ratio will most likely increase the intrinsic value when using a(n):

Correct answer: B

A. multiplier model. - Incorrect

Incorrect Answer Feedback: Incorrect because the increase in earnings affects the multiplier models, but the increase in the dividend payout ratio does not have any effect.

B. present value model. - Correct

Correct Answer Feedback: Correct because an increase in the dividend payout ratio will increase the cash expected to be distributed to shareholders. The dividend discount model is the present value of the cash expected to be distributed to shareholders. Therefore an increase in the dividend payout ratio will increase the intrinsic value in a present value model.

C. asset-based valuation model. - Incorrect

Incorrect Answer Feedback: Incorrect because the increase in the dividend payout ratio does not have an effect on the asset-based valuation model.

Question 31

If Investor 1 has a lower risk aversion coefficient than Investor 2, will Investor 2’s optimal portfolio have a higher expected return on the capital allocation line?

Correct answer: B

ncorrect Answer Feedback: Incorrect because it assumes that ROE = net profit margin × leverage (omitting asset turnover); or = 8% × 1.2 = 0.096. Substituting into the sustainable growth rate = retention rate × ROE = (1 - 0.10/0.15) × 0.096 = 0.0320 = 3.2%.

Figure 22: Source equation for question 29, option A

Correct Answer Feedback: Correct because the sustainable growth rate = retention rate × ROE, where retention rate = (1 – payout ratio), and the payout ratio is the percentage of earnings that the company pays out as dividends to shareholders, or (EPS – DPS) / EPS, and using the DuPont analysis ROE = net profit margin × total asset turnover × leverage. The retention rate = 1 – (0.10 / 0.15) = 1 – 0.667 = 0.3333. ROE = 8% × 1.5 × 1.2 = 0.144. Substituting into the sustainable growth rate formula: 0.3333 × 0.144 = 0.04799 ≈ 4.8%.

Figure 23: Source equation for question 29, option B

A. Yes - Incorrect

Incorrect Answer Feedback: Incorrect because Investor 2 has a higher risk aversion coefficient, which means a lower risk tolerance and a lower expected return on the capital allocation line.

B. No because Investor 2 has a lower risk tolerance - Correct

Correct Answer Feedback: Correct because Investor 2 has a higher risk aversion coefficient, which means a lower risk tolerance and a lower expected return on the capital allocation line.

C. No because Investor 2 has a higher risk tolerance - Incorrect

Incorrect Answer Feedback: Incorrect because Investor 2 has a higher risk aversion coefficient, which means a lower risk tolerance and a lower expected return on the capital allocation line.

Question 32

An analyst gathers the following information about a company's earnings:

Retention rate	40%
Growth rate	1%

According to the Gordon growth model, if an investor's required return is 6%, the justified forward P/E is:

Correct answer: C

A. 8. - Incorrect

Incorrect Answer Feedback: Incorrect but is the result when the retention rate is substituted for the payout rate: 0.4/(0.06 – 0.01) = 0.4/0.05 = 8.

B. 10. - Incorrect

Incorrect Answer Feedback: Incorrect because the growth rate is omitted: 0.6/(0.06) = 10.

C. 12. - Correct

Correct Answer Feedback: Correct because the dividend payout ratio = (1 – earnings retention rate) = (1 – 0.40) = 0.6. The justified forward P/E is: p/(r – g) where p = dividend payout ratio, r = required rate of return and g = sustainable growth rate, therefore P/E = 0.6/ (0.06 – 0.01) = 0.6/0.05 = 12.

Question 33

A hedge fund begins the year with $120 million and earns a 25% return for the year. The fund charges a 1.5% management fee on end-of-year fund value and a 15% incentive fee on the return, net of the management fees, that is in excess of a 6% fixed hurdle rate. The fund's investors' return for the year, net of fees, is closest to:

Correct answer: B

ncorrect Answer Feedback: Incorrect but is the result when the retention rate is substituted for the payout ate: 0.4/(0.06 – 0.01) = 0.4/0.05 = 8.

Figure 25: Source equation for question 32, option A

ncorrect Answer Feedback: Incorrect because the growth rate is omitted: 0.6/(0.06) = 10.

Figure 26: Source equation for question 32, option B

Correct Answer Feedback: Correct because the dividend payout ratio = (1 – earnings retention rate) = (1 – 0.40) = 0.6. The justified forward P/E is:

p/(r – g) where *p* = dividend payout ratio, *r* = required rate of return and *g* = sustainable growth rate, therefore P/E = 0.6/ (0.06 – 0.01) = 0.6/0.05 = 12.

answer)

Figure 27: Source equation for question 32, option C

ncorrect Answer Feedback: Incorrect because it fails to adjust for the hurdle rate when calculating the incentive fee or 0.15 × $27.75 million = $4.1625 million and $27.75 million – $4.1625 million = $23.5875 million. $23.5875 million/$120 million = 19.66%.

Figure 28: Source equation for question 33, option A

Figure 29: Source equation for question 33, option A

B. 20.56%. - Correct

Correct Answer Feedback: Correct because the $120 million grows by 25% to $150 million [= $120 million × (1 + 0.25)]. The management fee is $2.25 million (= $150 million × 0.015), leaving $147.75 million, net of the management fee, or an increase of $27.75 million over the beginning value of $120 million. The 6% hurdle rate requires an increase of $7.2 million (= $120 million × 0.06), so the fund has earned $20.55 million (= $27.75 million – $7.2 million) over the hurdle rate, net of the management fee. The incentive fee is 15% of this, or $3.0825 million (= $20.55 million × 0.015), leaving an increase in fund assets, net of management and incentive fees, of $24.6675 million (= $27.75 million – $3.0825 million). The investors' return, net of fees, is $24.6675/$120 million = 20.56%.

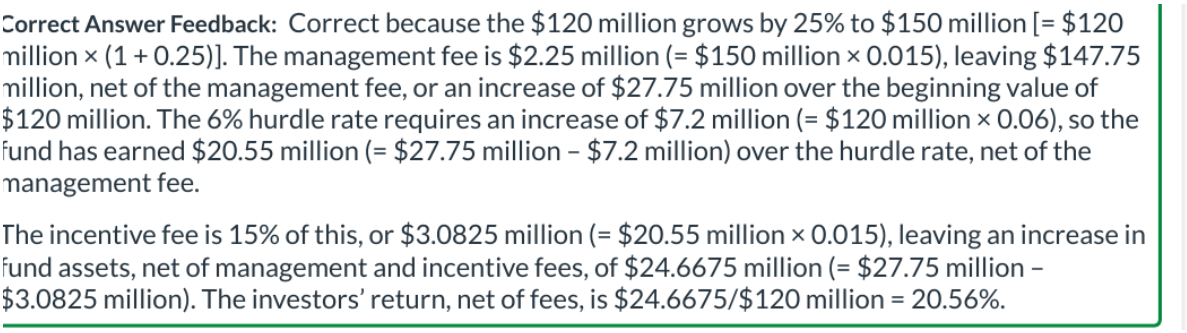


Figure 30: Source equation for question 33, option B

C. 21.25%. - Incorrect

Incorrect Answer Feedback: Incorrect because it ignores that management fee. The fund grows to $150 million. Incentive fee is ($150 million – $120 million) × 0.15 = $4.5 million $150 million – $120 million – $4.5 million = $25.5 million $25.5 million/$120 million = 0.2125

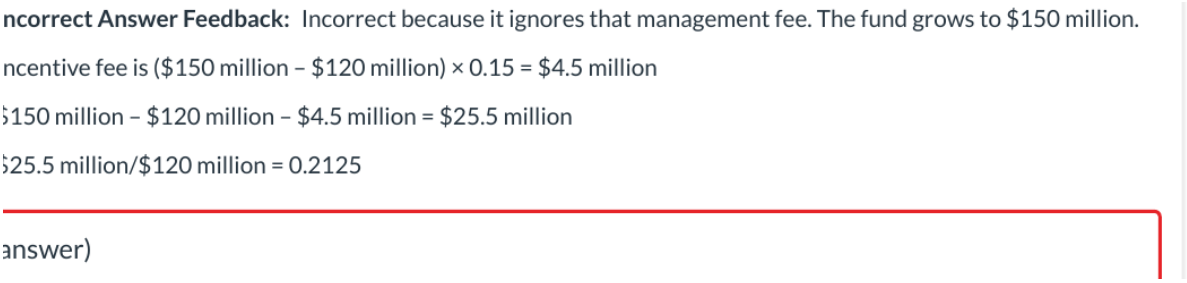


Figure 31: Source equation for question 33, option C

Question 34

All else being equal, which of the following bonds most likely has the highest yield to maturity?

Correct answer: C

A. Putable bonds - Incorrect

Incorrect Answer Feedback: Incorrect because a put provision gives the bondholders the right to sell the bond back to the issuer at a pre-determined price on specified dates. Because a put provision has value to the bondholders, the price of a putable bond will be higher than the price of an otherwise similar bond issued without the put provision. Similarly, the yield on a bond with a put provision will be lower than the yield on an otherwise similar non-putable bond.

B. Convertible bonds - Incorrect

Incorrect Answer Feedback: Incorrect because from the investor's perspective, a convertible bond offers several advantages relative to a non- convertible bond. Because the conversion provision is valuable to bondholders, the price of a convertible bond is higher than the price of an otherwise similar bond without the conversion provision. Similarly, the yield on a convertible bond is lower than the yield on an otherwise similar non-convertible bond.

C. Contingent convertible bonds - Correct

Correct Answer Feedback: Correct because bonds with contingent write-down provisions are convertible on the downside. In the case of contingent convertible bonds (CoCos), conversion is automatic if a specified event occurs. In such cases, the CoCos immediately convert into equity. Because the conversion is not at the option of the bondholders but automatic, CoCos force bondholders to take losses. For this reason, CoCos must offer a higher yield than otherwise similar bonds.

Question 35

According to behavioral finance, observed overreaction in securities markets most likely occurs because of:

Correct answer: A

A. loss aversion. - Correct

Correct Answer Feedback: Correct because according to loss aversion–related arguments in behavioral theories, investors dislike losses more than they like comparable gains. Thus, such a behavioral bias can explain observed overreaction in markets.

B. disposition effect. - Incorrect

Incorrect Answer Feedback: Incorrect because the disposition effect is a behavioral bias in which investors tend to avoid realizing losses but rather seek to realize gains.

C. gambler's fallacy. - Incorrect

Incorrect Answer Feedback: Incorrect because gambler's fallacy is a behavioral bias in which recent outcomes affect investors' estimates of future probabilities.

Question 36

Using put–call parity, a long call can best be replicated by going:

Correct answer: C

A. long the put, short the asset, and long the bond. - Incorrect

Incorrect Answer Feedback: Incorrect because the short asset position must be a long position, and the long bond position must be a short position. According to put–call parity, a long call is equal to long put, long asset, short bond.

B. short the put, long the asset, and short the bond. - Incorrect

Incorrect Answer Feedback: Incorrect because the short put position must be a long position. According to put–call parity, a long call is equal to long put, long asset, short bond.

C. long the put, long the asset, and short the bond. - Correct

Correct Answer Feedback: Correct because according to put–call parity, a long call is equal to long put, long asset, short bond.

Question 37

The value of a European call option is inversely related: The value of a European call option is inversely related:

Correct answer: B

A. only to the risk-free interest rate. - Incorrect

Incorrect Answer Feedback: Incorrect because the value of a European call is directly related to the risk-free interest rate. It is therefore not inversely related to the risk-free interest rate.

B. only to the dividends paid by the underlying stock. - Correct

Correct Answer Feedback: Correct because a European call option is worth less the more benefits that are paid by the underlying and worth more the more costs that are incurred in holding the underlying. Dividends are benefits paid by the underlying. So, a European call must be worth less when the underlying pays dividend than when it does not. Therefore, dividends paid by the underlying is inversely related to the value of a European call option.

C. both to the risk-free interest rate and to the dividends paid by the underlying stock. - Incorrect

Incorrect Answer Feedback: Incorrect because as described in the response rationale for the correct answer, the value of a European call option is inversely related to only the dividends paid by the underlying stock.

Question 38

With respect to index construction, which method of grouping companies most likely experiences the highest turnover in its constituents? Grouping by:

Correct answer: C

A. industry. - Incorrect

Incorrect Answer Feedback: Incorrect because company groupings based on statistical similarities tend to have greater constituent turnover than those based on industries or countries. These groupings are usually relative (e.g., companies with dividend yields at or above the 75th percentile as of the measurement date) and tend to show far more turnover in their constituents than groupings based on industries and countries, because these statistics are less stable by company and companies' rankings change.

B. geography. - Incorrect

Incorrect Answer Feedback: Incorrect because company groupings based on statistical similarities tend to have greater constituent turnover than those based on industries or countries. These groupings are usually relative (e.g., companies with dividend yields at or above the 75th percentile as of the measurement date) and tend to show far more turnover in their constituents than groupings based on industries and countries, because these statistics are less stable by company and companies' rankings change.

C. statistical similarities. - Correct

Correct Answer Feedback: Correct because company groupings based on statistical similarities tend to have greater constituent turnover than those based on industries or countries. These groupings are usually relative (e.g., companies with dividend yields at or above the 75th percentile as of the measurement date) and tend to show far more turnover in their constituents than groupings based on industries and countries, because these statistics are less stable by company and companies' rankings change.

Question 39

Unlike traditional financial assets, digital assets are most likely:

Correct answer: C

A. subject to well-defined regulatory standards. - Incorrect

Incorrect Answer Feedback: Incorrect because legal and regulatory standards for traditional financial instruments and assets and trading in these are well developed and are often comparable across national borders; this is not the case with digital assets. Comprehensive rules that are specific to digital assets—their formation, trading, and legal standards—are still being developed.

B. recorded in private ledgers maintained by central intermediaries. - Incorrect

Incorrect Answer Feedback: Incorrect because traditional assets are generally recorded in private ledgers maintained by central intermediaries. The ownership and exchange of digital assets are generally recorded on a decentralized digital ledger using cryptography and enhanced algorithms.

C. valued based on anticipated price appreciation rather than underlying cash flow. - Correct

Correct Answer Feedback: Correct because most digital assets do not have an inherent value based on underlying assets or on the potential cash flow—interest and dividends—they can or are expected to generate. They derive their price solely from an anticipated asset appreciation from a perceived scarcity value (due to limitations on the total supply of currency) and the potential ability to transfer value in the future (due to unique features in the underlying algorithms that may facilitate certain types of financial transactions).

Question 40

Which of the following transactions most likely represents a fair value hedge?

Correct answer: A

A. Commodity futures to hedge inventory - Correct

Correct Answer Feedback: Correct because it is an example of a fair value hedge. A fair value hedge designation applies when a derivative is deemed to offset the fluctuation in fair value of an asset or liability. For example, a commodities producer might sell its inventory forward in anticipation of lower future prices.

B. Currency forward to hedge forecasted sales - Incorrect

Incorrect Answer Feedback: Incorrect because it is an example of a cash flow hedge. Derivatives designated as absorbing the variable cash flow of a floating-rate asset or liability such as foreign exchange, interest rates, or commodities are referred to as cash flow hedges. For instance, the FX forward to hedge forecasted sales constitutes an example of a cash flow hedge.

C. Interest rate swap to a fixed rate for floating-rate debt - Incorrect

Incorrect Answer Feedback: Incorrect because it is an example of a cash flow hedge. Derivatives designated as absorbing the variable cash flow of a floating-rate asset or liability such as foreign exchange, interest rates, or commodities are referred to as cash flow hedges. A swap to a fixed rate for a floating-rate debt liability is another example of a cash flow hedge.

Question 41

If the current share price is $60, a trader who wants to purchase the share at $58 or less will most likely place a:

Correct answer: A

A. limit order. - Correct

Correct Answer Feedback: Correct because a limit order conveys almost the same instruction as market order: Obtain the best price immediately available, but in no event accept a price higher than a specified limit price when buying or accept a price lower than a specified limit price when selling.

B. market order. - Incorrect

Incorrect Answer Feedback: Incorrect because a market order instructs the broker or exchange to obtain the best price immediately available when filling the order. The trader might purchase it at more than $58 by using market order.

C. stop-buy order. - Incorrect

Incorrect Answer Feedback: Incorrect because a stop-buy order is to buy at a price specified above the market price. Because stop-sell orders become valid when prices are falling and stop-buy orders become valid when prices are rising.

Question 42

A lifecycle fund will most likely:

Correct answer: C

A. maintain a fixed asset allocation of stocks and bonds. - Incorrect

Incorrect Answer Feedback: Incorrect because the asset allocation would change as described in the response rationale for the answer key.

B. use tactical asset allocation to capture market opportunities. - Incorrect

Incorrect Answer Feedback: Incorrect because the asset allocation will change as described in the response rationale for the answer key, but will change due to target date, not due to market opportunities. Tactical asset allocation is a short-term change in allocation to capture market opportunities.

C. change the asset allocation as the fund nears its target date. - Correct

Correct Answer Feedback: Correct because lifecycle or Target Date funds manage the asset mix based on a desired retirement date. For example, a Target Date fund 40 years into the future might be 90 percent invested in equities and 10 percent in fixed income. As time passes, however, the fund would gradually change the mix of shares from equity to fixed income.

Question 43

A trader enters into the following margin transaction:

Purchase price, per share	€100
Initial margin posted as equity	45%
Holding period	1 year
Unleveraged security return during the period	15%
Call money rate, one year	4%

If there are no dividends or commissions, and interest on the margin loan is paid at the end of the period, the trader's rate of return on the margin transaction is closest to:

Correct answer: B

A. 27.3%. - Incorrect

Incorrect Answer Feedback: Incorrect because it is the unleveraged return divided by 1 – the margin requirement = 0.15 / (1 – 0.45) = 0.2727 ≈ 27.3%.

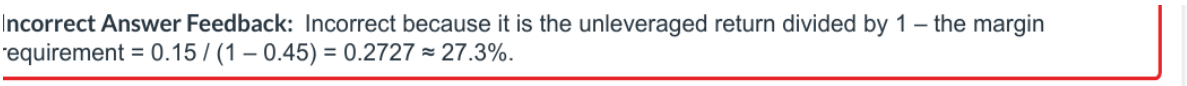


Figure 32: Source equation for question 43, option A

B. 28.4%. - Correct

Correct Answer Feedback: Correct because the initial margin requirement is the minimum fraction of the purchase price that must be trader's equity. With the purchase price = $100, initial investment = purchase price × initial margin requirement = €100 × 45% = €45.00; trading gains/losses = purchase price × unleveraged security return = €100 × 15% = €15.00; margin interest paid = purchase price × (1−initial margin requirement) × call money rate = $100 × (1−45%) × 4% = €2.20. Remaining equity = initial investment + trading gains/losses – margin interest paid. Substituting, remaining equity = €45.00 + €15.00 − €2.20 = €57.80. Rate of return = (remaining equity − initial investment) / initial investment. Substituting, rate of return = (€57.80 − €45.00) / €45.00 = 0.2844 or 28.4%.

C. 29.3%. - Incorrect

Incorrect Answer Feedback: Incorrect because margin interest paid is calculated based on the initial investment. Margin interest paid = initial investment × call money rate = €45.00 × 4% = €1.80. Remaining equity = €45.00 + €15.00 − €1.80 = €58.20. Rate of return = (€58.20 − €45.00) / €45.00 = 0.2933 ≈ 29.3%.

Correct Answer Feedback: Correct because the initial margin requirement is the minimum fraction of the purchase price that must be trader's equity. With the purchase price = $100, initial investment = purchase price × initial margin requirement = €100 × 45% = €45.00; trading gains/losses = purchase price × unleveraged security return = €100 × 15% = €15.00; margin interest paid = purchase price × (1−initial margin requirement) × call money rate = $100 × (1−45%) × 4% = €2.20. Remaining equity = initial investment + trading gains/losses – margin interest paid. Substituting, remaining equity = €45.00 + €15.00 − €2.20 = €57.80. Rate of return = (remaining equity − initial investment) / initial investment. Substituting, rate of return = (€57.80 − €45.00) / €45.00 = 0.2844 or 28.4%.

Figure 33: Source equation for question 43, option B

Incorrect Answer Feedback: Incorrect because margin interest paid is calculated based on the initial investment. Margin interest paid = initial investment × call money rate = €45.00 × 4% = €1.80. Remaining equity = €45.00 + €15.00 − €1.80 = €58.20. Rate of return = (€58.20 − €45.00) / €45.00 = 0.2933 ≈ 29.3%.

Figure 34: Source equation for question 43, option C

Question 44

In which form of market efficiency do security prices fully reflect private information?

Correct answer: C

A. Weak-form efficient - Incorrect

Incorrect Answer Feedback: Incorrect because weak-form efficiency makes no claims about prices reflecting private information, only that prices reflect all past market data such as price and volume information.

B. Semi-strong-form efficient - Incorrect

Incorrect Answer Feedback: Incorrect because semi-strong form efficiency makes no claims on prices reflecting private information, only that prices reflect all past market data such as price and volume information and all publicly known information such as financial statements.

C. Strong-form efficient - Correct

Correct Answer Feedback: Correct because in a strong-form efficient market, security prices fully reflect private information. private information.

Question 45

The data for two stocks in an index are as follows: Percent of Shares in Beginning of Period End of Period Dividends per Stock Shares Outstanding Market Float Price ($) Price ($) Share ($) A 5,000 90 40 45 1.00 B 2,000 100 68 60 0.50 Assuming the beginning value of the float-adjusted market-capitalization-weighted equity index is 100, the ending value is closest to:

Correct answer: A

A. 102.06. - Correct

Correct Answer Feedback: Correct. In float-adjusted market-capitalization weighting, the weight on each constituent security is determined by adjusting its market capitalization for its market float. As the following computations show, the ending value of the index equals 102.06 (322,500/316,000). Percent of Beginning Ending Beginning of End of Shares Shares in Shares in Float Adj. Float Adj. Period Price Period Stock Outstanding Market Index Market Cap Market Cap ($) Price ($) (1) Float (1 × 2 = 3) ($) ($) (4) (6) (2) (3 × 4 = 5) (3 × 6) A 5,000 90 4,500 40 180,000 45 202,500 B 2,000 100 2,000 68 136,000 60 120,000 Total 316,000 322,500 Index value 100 102.06

Correct Answer Feedback: Correct. In float-adjusted market-capitalization weighting, the weight on each constituent security is determined by adjusting its market capitalization for its market float. As the following computations show, the ending value of the index equals 102.06 (322,500/316,000).

Stock	Shares Outstanding (1)	Percent of Shares in Market Float (2)	Shares in Index (1 × 2 = 3)	Beginning of Period Price ($) (4)	Beginning Float Adj. Market Cap ($) (3 × 4 = 5)	End of Period Price ($) (6)	Ending Float Adj. Market Cap ($) (3 × 6)
A	5,000	90	4,500	40	180,000	45	202,500
B	2,000	100	2,000	68	136,000	60	120,000
Total					316,000		322,500
Index value					100		102.06

Figure 35: Source equation for question 45, option A

B. 102.68. - Incorrect

Incorrect Answer Feedback: Incorrect. It ignores the float and simply computes the index on the basis of market-capitalization weighting. Shares Beginning of Period Beg. Float Adj. End of Period Ending Float Adj. Stock Outstanding Price ($) Market Cap ($) Price ($) Market Cap ($) A 5,000 40 200,000 45 225,000 B 2,000 68 136,000 60 120,000 Total 336,000 345,000 Index value 100 102.68

C. 103.80. - Incorrect

Incorrect Answer Feedback: Incorrect. It does make adjustment for the market float but makes the mistake of including dividends in the ending price of the stock. % Shares in Shares Beginning of End of Period Ending Float Shares Beg. Float Adj. Stock Market in Period Price Price + Dividend Adj. Market Cap Outstanding Market Cap ($) Float Index ($) ($) ($) A 5,000 90 4,500 40 180,000 46 207,000 B 2,000 100 2,000 68 136,000 60.5 121,000 Total 316,000 328,000 Index value 100 103.80

Question 46

A trader sells a put option with an exercise price of $33.50 for $3.00. If the price of the underlying asset at expiration is $29.50, the trader's profit is:

Correct answer: A

A. –$1.00. - Correct

Correct Answer Feedback: Correct because the profit to the put seller is Π = –Max(0,X – S) + p , where Π is the T 0 profit, X is the exercise price, S is the price of the underlying asset at expiration, and p is the price of the put T 0 option at initiation or the purchase (or sale) price of the put option. Thus, the profit is Π = –Max(0,$33.50 – $29.50) + $3.00 = –$4.00 + $3.00 = –$1.00.

Correct Answer Feedback: Correct because the profit to the put seller is $\Pi = -\text{Max}(0, X - S_T) + p_0$, where Π is the profit, X is the exercise price, S_T is the price of the underlying asset at expiration, and p_0 is the price of the put option at initiation or the purchase (or sale) price of the put option. Thus, the profit is $\Pi = -\text{Max}(0,\$33.50 - \$29.50) + \$3.00 = -\$4.00 + \$3.00 = -\$1.00$.

Figure 36: Source equation for question 46, option A

B. $1.00. - Incorrect

Incorrect Answer Feedback: Incorrect because $1.00 represents the profit to the buyer of the put option, not the profit to the seller. The profit to the put buyer" is Π = Max(0,X – S) – p , where Π is the profit, X is the exercise T 0 price, S is the price of the underlying asset at expiration, and p is the price of the put option at initiation or the T 0 purchase (or sale) price of the put option. Thus, the profit to the put buyer is Π = Max(0,$33.50 – $29.50) – $3.00 = $4.00 – $3.00 = $1.00.

ncorrect Answer Feedback: Incorrect because $1.00 represents the profit to the buyer of the put option, not he profit to the seller. The profit to the put buyer" is $\Pi = \text{Max}(0, X - S_T) - p_0$, where Π is the profit, X is the exercise rice, S_T is the price of the underlying asset at expiration, and p_0 is the price of the put option at initiation or the urchase (or sale) price of the put option. Thus, the profit to the put buyer is $\Pi = \text{Max}(0, \$33.50 - \$29.50) - \$3.00 = \$4.00 - \$3.00 = \$1.00$.

Figure 37: Source equation for question 46, option B

C. $4.00. - Incorrect

Incorrect Answer Feedback: Incorrect because it is the difference between the exercise price X and the price of the underlying asset at expiry S , or $33.50 – $29.50 = $4.00, which is equal to the value of this in-the-money put T at expiration for the put buyer. It is not equal to the profit for the put seller of –$1.00.

ncorrect Answer Feedback: Incorrect because it is the difference between the exercise price X and the price of he underlying asset at expiry S_T, or $33.50 – $29.50 = $4.00, which is equal to the value of this in-the-money put t expiration for the put buyer. It is not equal to the profit for the put seller of –$1.00.

answer)

Figure 38: Source equation for question 46, option C

Question 47

A bond has a 10-year maturity, a $1,000 face value, and a 7% coupon rate. If the market requires a yield of 8% on similar bonds, it will most likely trade at a:

Correct answer: A

A. discount. - Correct

Correct Answer Feedback: Correct because when the required yield is higher than the coupon rate, the bond will trade at a discount to par.

B. premium. - Incorrect

Incorrect Answer Feedback: Incorrect because a bond trades at a premium when the required yield is less than the coupon rate.

C. discount or premium, depending on its duration. - Incorrect

Incorrect Answer Feedback: Incorrect because a bond trades at a discount when the required yield is higher than the coupon rate.

Question 48

A bond has an annual modified duration of 8 and an annual convexity measure of 150. If the bond's yield to maturity decreases by 140 bps, the change in the price of this bond is closest to:

Correct answer: B

A. 9.7%. - Incorrect

Incorrect Answer Feedback: Incorrect because the second term is subtracted instead of added: Expected price $\Delta \neq [-8 \times (-0.0140)] - [0.5 \times (150) \times (-0.0140)2] = 0.112 - 0.0147 = 0.0973 = 9.7\%$.

B. 12.7%. - Correct

Correct Answer Feedback: Correct because: Expected price change = (− AnnModDur × Δyield) + [0.5 × AnnConvexity × (Δyield)2] = [− 8 × (− 0.0140)] + [0.5 × (150) × (− 0.0140)2] = 0.112 + 0.0147 = 0.1267 = 12.7%.

ncorrect Answer Feedback: Incorrect because the second term is subtracted instead of added: Expected price Δ

$- 8 \times (-0.0140)] - [0.5 \times (150) \times (-0.0140)^2] = 0.112 - 0.0147 = 0.0973 = 9.7\%.$

Figure 39: Source equation for question 48, option A

Correct Answer Feedback: Correct because:

Expected price change = (− AnnModDur × Δyield) + [0.5 × AnnConvexity × (Δyield)^2] =

$- 8 \times (-0.0140)] + [0.5 \times (150) \times (-0.0140)^2] = 0.112 + 0.0147 = 0.1267 = 12.7\%.$

Figure 40: Source equation for question 48, option B

C. 14.1%. - Incorrect

Incorrect Answer Feedback: Incorrect because “0.5” is omitted from the second term of calculation: Expected price Δ ≠ [− 8 × (− 0.0140)] + [(150) × (− 0.0140)2] = 0.112 + 0.0294 = 0.1414 = 14.1%.

ncorrect Answer Feedback: Incorrect because “0.5” is omitted from the second term of calculation: Expected
rice Δ ≠

$- 8 \times (-0.0140)] + [(150) \times (-0.0140)^2] = 0.112 + 0.0294 = 0.1414 = 14.1\%.$

answer)

Figure 41: Source equation for question 48, option C

Question 49

At expiration, an option that is in the money will most likely have:

Correct answer: B

A. time value, but no exercise value. - Incorrect

Incorrect Answer Feedback: Incorrect. At expiration, options have no time value.

B. exercise value, but no time value. - Correct

Correct Answer Feedback: Correct. At expiration, options have no time value; if they are in the money, they have exercise value.

C. both time value and exercise value. - Incorrect

Incorrect Answer Feedback: Incorrect. At expiration, options have no time value.

Question 50

Investing in an existing solar power plant is most appropriately categorized as a:

Correct answer: A

A. utility asset. - Correct

Correct Answer Feedback: Correct because utility investments encompass environmentally sustainable development, with an increasing focus on renewable technologies, including solar, wind, and waste-to-energy power generation.

B. greenfield investment. - Incorrect

Incorrect Answer Feedback: Incorrect because a newly constructed solar power plant is a brownfield investment instead of a greenfield investment. Infrastructure investments may also be categorized by the underlying asset's stage of development. Greenfield investments are defined as developing new assets and new infrastructure. By contrast, brownfield investments expand existing facilities and may involve privatization of public assets or a sale leaseback of completed greenfield projects.

C. social infrastructure asset. - Incorrect

Incorrect Answer Feedback: Incorrect because a newly constructed solar power plant is a utility asset instead of a social infrastructure asset. Social infrastructure investments are directed toward human activities and include such assets as educational, health care, social housing, and correctional facilities, with the focus on providing, operating, and maintaining the asset infrastructure.

Question 51

An analyst gathers the following information about three companies and their common shares:

	Company 1	Company 2	Company 3
Next year's expected dividend (D) 1	$1.00	$1.00	$1.25
Dividend payout ratio	10%	20%	15%

All three companies have the same ROE of 10% and required return of 10%. Using the Gordon growth model, the company with the highest intrinsic value is:

Correct answer: A

A. Company 1. - Correct

Correct Answer Feedback: Correct because the estimated value (V) for Company 1 is highest, at $100 / share: 0 V = D ÷(r-g) 0 1 Where: D = next year's expected dividend 1 r = required return g = dividend growth rate g = b × ROE b = earnings retention rate = 1 – dividend payout ratio b = (1 – 0.10) = 0.90; g = 0.90 × 0.10 = 0.09 V = 1.00 / (0.10 – 0.09) = 1.00 / 0.01 = $100.

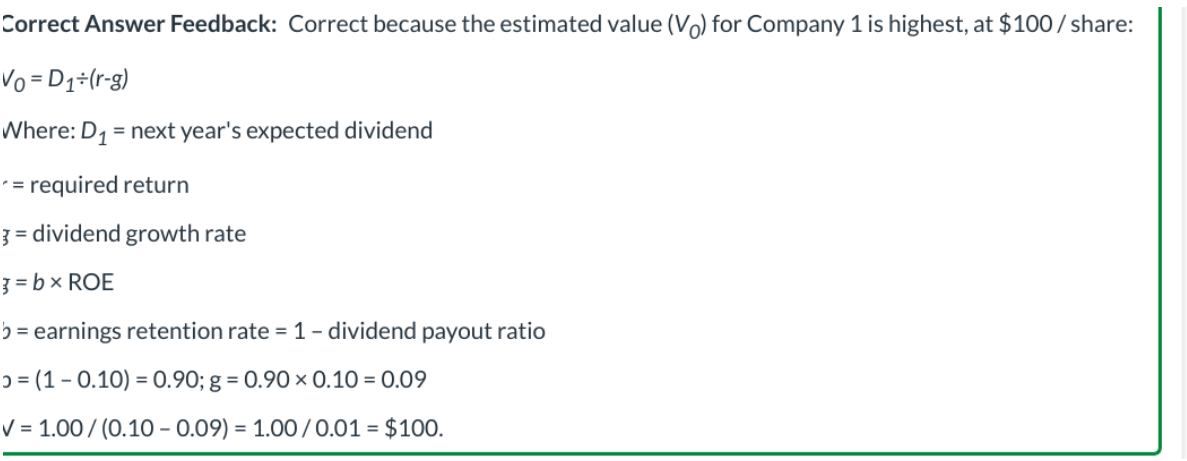


Figure 42: Source equation for question 51, option A

B. Company 2. - Incorrect

Incorrect Answer Feedback: Incorrect because the value of Company 2 is lower than that of Company 1 at $50 / share: b = (1 – 0.20) = 0.80; g = 0.80 × 0.10 = 0.08 V = 1.00 / (0.10 – 0.08) = 1.00 / 0.02 = $50.00. A candidate may also mistakenly choose this answer choice because it has the highest dividend payout ratio.

ncorrect Answer Feedback: Incorrect because the value of Company 2 is lower than that of Company 1 at $50 / hare:

Figure 43: Source equation for question 51, option B

) = (1 – 0.20) = 0.80; g = 0.80 × 0.10 = 0.08

/ = 1.00 / (0.10 – 0.08) = 1.00 / 0.02 = $50.00. A candidate may also mistakenly choose this answer choice ecause it has the highest dividend payout ratio.

Figure 44: Source equation for question 51, option B

C. Company 3. - Incorrect

Incorrect Answer Feedback: Incorrect because the value of Company 3 is lower than that of Company 1 at $83.33 / share: b = (1 – 0.15) = 0.85; g = 0.85 × 0.10 = 0.085 V = 1.25 / (0.10 – 0.085) = 1.25 / 0.015 ≈ $83.33. A candidate may also mistakenly choose this answer choice because it has the highest dividend.

ncorrect Answer Feedback: Incorrect because the value of Company 3 is lower than that of Company 1 at 83.33 / share:

) = (1 – 0.15) = 0.85; g = 0.85 × 0.10 = 0.085

/ = 1.25 / (0.10 – 0.085) = 1.25 / 0.015 ≈ $83.33. A candidate may also mistakenly choose this answer choice ecause it has the highest dividend.

answer)

Figure 45: Source equation for question 51, option C

Question 52

With respect to the capital market line, the market portfolio is best described as the:

Correct answer: A

A. optimal risky portfolio. - Correct

Correct Answer Feedback: Correct because the specific capital allocation line that uses the market portfolio as the optimal risky portfolio is known as the capital market line (CML). Further, theoretically, the market includes all risky assets. Because the CML assumes homogeneous expectations, only one optimal (risky) portfolio exists.

B. portfolio that maximizes an investor's utility. - Incorrect

Incorrect Answer Feedback: Incorrect because an investor's optimal portfolio (the combination of the risk-free asset and the market portfolio that maximizes an individual investor's utility) could be any point on the CML. The location of an optimal investor portfolio (not the optimal risky portfolio) depends on the investor's risk preferences. A highly risk-averse investor may invest a large proportion, even 100 percent, of his/her assets in the risk-free asset. A less risk-averse investor, however, may invest a large portion of his/her wealth in the optimal risky asset.

C. optimal combination of the risk-free asset and a portfolio of risky assets. - Incorrect

Incorrect Answer Feedback: Incorrect because the market portfolio is the optimal risky portfolio rather than a combination of the risk-free asset and a portfolio of risky assets. All combinations of the risk-free asset and a risky asset lie on the capital allocation line. Superimposing an investor's indifference curves on the capital allocation line will lead to the optimal investor portfolio. An investor's optimal portfolio could be any point on the CML.

Question 53

Which of the following statements is most accurate? In derivatives pricing:

Correct answer: B

A. investors are assumed to be risk averse. - Incorrect

Incorrect Answer Feedback: Incorrect because the important point to understand is that while the risk aversion of investors is relevant to pricing assets, it is not relevant to pricing derivatives. As such, derivatives pricing is sometimes called risk-neutral pricing. Therefore, investors are not assumed to be risk averse. B. expected payoffs of the derivative can be discounted at the risk-free rate.

B. expected payoffs of the derivative can be discounted at the risk-free rate. - Correct

Correct Answer Feedback: Correct because the risk aversion of the investor is not relevant to pricing the derivative, one can just as easily obtain the derivative price by assuming that the investor is risk neutral. That means that the expected payoff of the derivative can be discounted at the risk-free rate rather than the risk-free rate plus a risk premium.

C. a portfolio consisting of the underlying and the derivative must earn the risk-free rate plus a risk premium. - Incorrect

Incorrect Answer Feedback: Incorrect because the derivative price is the price that guarantees the risk-free combination of the derivative and the underlying produces a risk-free return. Also, risk-neutral pricing uses the fact that arbitrage opportunities guarantee that a risk-free portfolio consisting of the underlying and the derivative must earn the risk-free rate, not a rate of return equal to the risk-free rate plus a risk premium.

Question 54

An example of a negative bond covenant is a:

Correct answer: B

A. promise to pay required taxes. - Incorrect

Incorrect Answer Feedback: Incorrect because affirmative covenants are typically administrative in nature and include a promise to comply with all laws and regulations, maintain its current lines of business, insure and maintain its assets, and pay taxes as they come due.

B. limit on the issuance of additional debt. - Correct

Correct Answer Feedback: Correct because examples of negative covenants include the Restrictions on debt regulate the issue of additional debt.

C. requirement to comply with existing laws and regulations. - Incorrect

Incorrect Answer Feedback: Incorrect because affirmative covenants are typically administrative in nature and include a promise to comply with all laws and regulations, maintain its current lines of business, insure and maintain its assets, and pay taxes as they come due.

Question 55

The slope of the security market line is best derived from the:

Correct answer: B

A. beta of the security. - Incorrect

Incorrect Answer Feedback: Incorrect because the beta of the security is shown on the x-axis.

B. market risk premium. - Correct

Correct Answer Feedback: Correct because the security market line is a graphical representation of the CAPM with beta on the x-axis and expected return on the y-axis. The slope of the line is given by the market risk premium, the difference between the equity market return and the risk-free rate of interest.

C. risk-free rate of return. - Incorrect

Incorrect Answer Feedback: Incorrect because the risk-free rate of return marks the intercept term of the security market line.

Question 56

Which of the following statements is most accurate?

Correct answer: A

A. The value of a swap fluctuates as market conditions change - Correct

Correct Answer Feedback: Correct because the value of a swap will fluctuate as market conditions change.

B. The price and the value of a swap are the same at expiration - Incorrect

Incorrect Answer Feedback: Incorrect because the world of forwards, futures, and swaps, however, uses different terminology with respect to price and value. As the underlying moves, swap values become either positive or negative. The forward, futures, or swap price is a concept that represents the fixed price or rate at which the underlying will be purchased at a later date. The value and price are not at all comparable with each other. Therefore, the statement that the price and the value of a swap are the same is incorrect.

C. The price of a swap at initiation is zero as there is no initial outlay of cash - Incorrect

Incorrect Answer Feedback: Incorrect because it is the value of a swap contract that is zero at initiation and not the price. Forwards, futures, and swaps start off with values of zero.

Question 57

In a declining interest rate environment, an investor would find it most difficult to estimate future cash flows for which of the following bonds with similar maturities?

Correct answer: C

A. Putable bond - Incorrect

Incorrect Answer Feedback: Incorrect because if interest rates decline, a putable bond will increase in value to a level above the put price and the bondholder will not put the bond to the issuer (long the option).

B. Floating-rate bond with a floor - Incorrect

Incorrect Answer Feedback: Incorrect because even though coupon payments will decline as interest rates decline, the floor sets a minimum (and constant) level of payments when rates decline to and below the floor level. In this sense, forecasting cash flows is less difficult than for the pass-through security.

C. Mortgage pass-through security - Correct

Correct Answer Feedback: Correct because when interest rates decline, borrowers will accelerate their prepayments. Compared to the floater with a floor, the mortgage pass-through has no restriction on the dollar amount of the prepayments. Prepayments are modeled to a benchmark but actual experience may be greater or less than forecast.

Question 58

An analyst gathers the following information on three investors. Each investor holds a bond with a Macaulay duration of 5.5 years in his portfolio:

	Investment Horizon
Investor A	5 years
Investor B	2 years

	Investment Horizon
Investor C	8 years

All else equal, which investor is currently most vulnerable to an increase in interest rates?

Correct answer: B

A. Investor A - Incorrect

Incorrect Answer Feedback: Incorrect because although Investor A has an investment horizon shorter than the Macaulay duration of 5.5 years, it is not substantially shorter. Also, he is not as vulnerable to an increase in rates as Investor B, whose bond has considerable market price risk.

B. Investor B - Correct

Correct Answer Feedback: Correct because Investor B has an investment horizon shorter than the Macaulay duration of 5.5 years. When the investment horizon is shorter than the Macaulay duration of the bond, market price risk dominates coupon reinvestment risk. The investor is vulnerable to higher interest rates.

C. Investor C - Incorrect

Incorrect Answer Feedback: Incorrect because Investor C has an investment horizon longer than the Macaulay duration. In this case, coupon reinvestment risk dominates market price risk. Also, assuming an interest rate increase, the gain from reinvesting coupon payments at a higher rate will outweigh any loss incurred from a drop in bond price.

Question 59

The sensitivity of a bond's price to a change in the benchmark yield curve is best described as:

Correct answer: A

A. effective duration. - Correct

Correct Answer Feedback: Correct because effective duration is a type of curve duration that measures the sensitivity of the bond price with respect to a benchmark yield curve.

B. modified duration. - Incorrect

Incorrect Answer Feedback: Incorrect because modified duration is a type of yield duration (as opposed to curve duration), which measures the sensitivity of a bond's price to a change in its own yield to maturity (not the benchmark yield curve).

C. the price value of a basis point. - Incorrect

Incorrect Answer Feedback: Incorrect because the price value of a basis point is a type of yield duration (as opposed to curve duration), which measures the sensitivity of a bond's price to a change in its own yield to maturity (not the benchmark yield curve).

Question 60

An analyst gathers the following information about a bond:

Market Price	96.00
Market price if yields rise 1 basis point	95.92
Market price if yields fall 1 basis point	96.10

The price value of a basis point is:

Correct answer: B

A. 0.08. - Incorrect

Incorrect Answer Feedback: Incorrect because only the increase in yield side is used to calculate PVBP. PVBP ≠ (PV – PV) + ≠ (96 – 95.92) = 0.08.

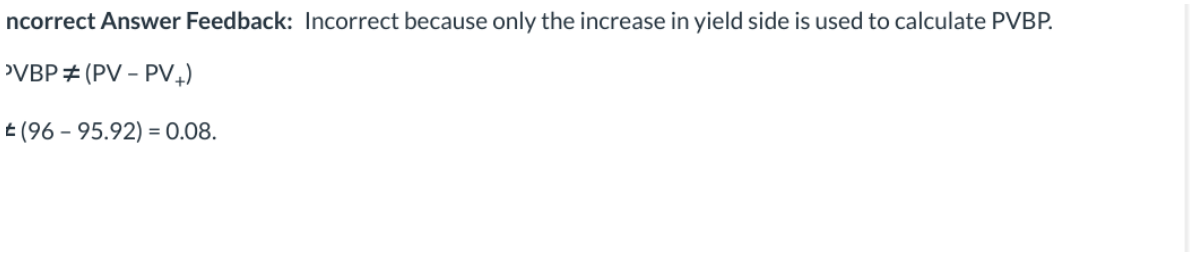


Figure 46: Source equation for question 60, option A

B. 0.09. - Correct

Correct Answer Feedback: Correct because the price value of a basis point (or PVBP) is an estimate of the change in the full bond price given a 1 bp change in the yield-to-maturity. PVBP = (PV – PV)/2 – + = (96.1 – 95.92)/2 = 0.09.

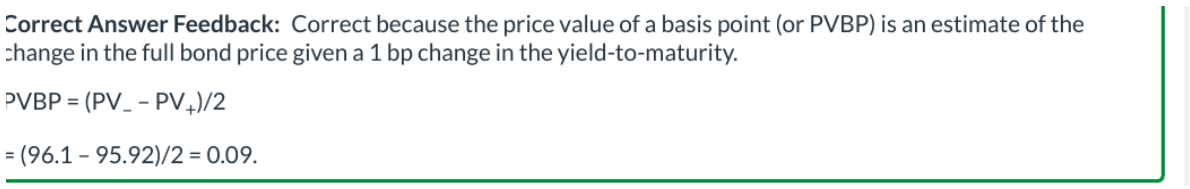


Figure 47: Source equation for question 60, option B

C. 0.18. - Incorrect

Incorrect Answer Feedback: Incorrect because the ‘divide by 2’ is missing in the formula. ≠ (96.1 – 95.92) = 0.18.

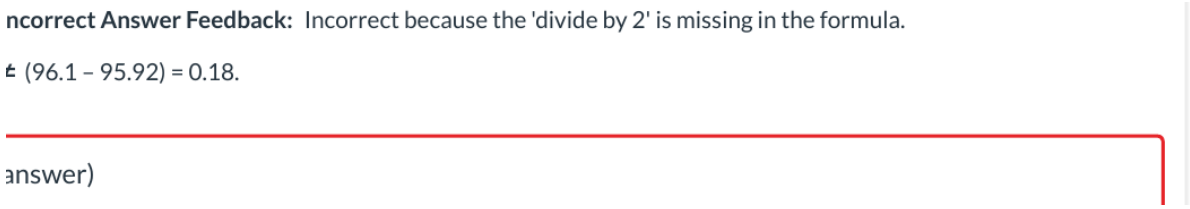


Figure 48: Source equation for question 60, option C

Question 61

The convexity of putable bonds is:

Correct answer: A

A. always positive. - Correct

Correct Answer Feedback: Correct. Putable bonds always have positive convexity.

B. negative for a decrease in yield. - Incorrect

Incorrect Answer Feedback: Incorrect because putable bonds always have positive convexity.

C. negative for an increase in yield. - Incorrect

Incorrect Answer Feedback: Incorrect because putable bonds always have positive convexity.

Question 62

Which of the following is used by clearinghouses to determine the daily gain and loss to the parties of a futures contract?

Correct answer: A

A. Mark to market - Correct

Correct Answer Feedback: Correct because the most important feature of futures contracts is the daily settlement of gains and losses and the associated credit guarantee provided by the exchange through its clearinghouse. At the end of each day, the clearinghouse engages in a practice called mark to market (MTM), also known as the daily settlement. The clearinghouse determines an average of the final futures trading price of the day and designates that price as the end-of-day settlement price. All contracts are then said to be marked to the end-of-day settlement price.

B. Maintenance margin - Incorrect

Incorrect Answer Feedback: Incorrect because the maintenance margin is the amount of money that each participant must maintain in the account after the trade is initiated, and it is always significantly lower than the initial margin. Therefore, the maintenance margin covers any potential losses arising from the mark to market practice which has to occur first before any shortfall in the maintenance margin is calculated.

C. Limit up and limit down - Incorrect

Incorrect Answer Feedback: Incorrect because some futures contracts contain a provision limiting price changes. These rules, called price limits, establish a band relative to the previous day's settlement price, within which all trades must occur. If market participants wish to trade at a price above the upper band, trading stops, which is called limit up, until two parties agree on a trade at a price lower than the upper limit. Likewise, if market participants wish to trade at a price below the lower band, which is called limit down, no trade can take place until two parties agree to trade at a price above the lower limit. Therefore, limit up and limit down refer to price limits but not to the practice that determines current gain and loss positions of futures contract parties.

Question 63

In the private debt market, a hybrid loan structure that combines secured and unsecured debt into a single loan with a blended interest rate is best described as:

Correct answer: A

A. unitranche debt. - Correct

Correct Answer Feedback: Correct because this is the definition of unitranche debt. Another type of debt that could be directly extended to borrowers is unitranche debt. Unitranche debt consists of a hybrid or blended loan structure combining different tranches of secured and unsecured debt into a single loan with a single, blended interest rate. Since unitranche debt is a blend of secured and unsecured debt, its interest rate will generally fall in between the interest rates often demanded on secured and unsecured debt. The unitranche loan will usually be structured between senior and subordinated debt in priority ranking.

B. mezzanine debt. - Incorrect

Incorrect Answer Feedback: Incorrect because mezzanine debt is not a blend of secured and unsecured debt. Mezzanine debt refers to private credit subordinated to senior secured debt but senior to equity in the borrower's capital structure. Because of its typically junior ranking and its usually unsecured status, mezzanine borrower's capital structure. Because of its typically junior ranking and its usually unsecured status, mezzanine debt is riskier than senior secured debt.

C. a leveraged loan. - Incorrect

Incorrect Answer Feedback: Incorrect because a leveraged loan is a type of direct lending that typically does not blend secured and unsecured debt. The debt itself typically is senior and secured and has covenants in place to protect the lender/investor. In direct lending, many firms may also provide debt in the form of a leveraged loan, a

loan that is itself levered. Private debt firms that invest in leveraged loans first borrow money to finance the debt and then extend it to another borrower.

Question 64

The two-fund separation theorem states that all investors will hold a combination of the:

Correct answer: A

A. risk-free asset and the optimal risky portfolio. - Correct

Correct Answer Feedback: Correct because the two-fund separation theorem states that all investors regardless of taste, risk preferences, and initial wealth will hold a combination of two portfolios or funds: a risk- free asset and an optimal portfolio of risky assets.

B. risk-free asset and the global minimum-variance portfolio. - Incorrect

Incorrect Answer Feedback: Incorrect because the two-fund separation theorem states that all investors regardless of taste, risk preferences, and initial wealth will hold a combination of two portfolios or funds: a risk- free asset and an optimal portfolio of risky assets. The left-most point on the minimum-variance frontier is the portfolio with the minimum variance among all portfolios of risky assets, and is referred to as the global minimum- variance portfolio. While the investor will hold the risk-free asset in their portfolio, their risk-return trade-off can be improved by holding the optimal risky portfolio rather than the global minimum-variance portfolio.

C. optimal risky portfolio and the global minimum-variance portfolio. - Incorrect

Incorrect Answer Feedback: Incorrect because the two-fund separation theorem states that all investors regardless of taste, risk preferences, and initial wealth will hold a combination of two portfolios or funds: a risk- free asset and an optimal portfolio of risky assets. The left-most point on the minimum-variance frontier is the portfolio with the minimum variance among all portfolios of risky assets, and is referred to as the global minimum- variance portfolio. While the investor will hold the optimal risky portfolio, their risk return trade-off can be improved by holding the risk-free asset rather than the global minimum-variance portfolio.

Question 65

The Gordon growth model is most appropriate for valuing the common stock of a dividend paying company that is:

Correct answer: B

A. young and just entering the growth phase. - Incorrect

Incorrect Answer Feedback: Incorrect. A three-stage dividend discount model would be most appropriate for a fairly young company that is just entering the growth phase.

B. mature and relatively insensitive to economic fluctuations. - Correct

Correct Answer Feedback: Correct. The Gordon growth model is most appropriate for valuing common stock of a dividend paying company that is mature and relatively insensitive to the business cycle or economic fluctuations. fluctuations.

C. experiencing growth that is higher than the sustainable growth rate. - Incorrect

Incorrect Answer Feedback: Incorrect. A two-stage dividend discount model would be appropriate for a company that is experiencing a higher than the sustainable growth rate.

Question 66

An analyst gathers the following balance sheet information (in $ millions) for a company:

Cash	2.5
Accounts receivable	7.5
Inventories	15.0
Net fixed assets	25.0
Total assets	50.0
Total liabilities	22.5
Common shareholders' equity	27.5
Total liabilities and equity	50.0

The market value of inventories is 110% of reported value and the market value of net fixed assets is 115% of reported value. All other values are as reported. There are 2.0 million shares outstanding and the current share price is $18.00. Using an asset-based valuation approach, the company's shares are most likely:

Correct answer: C

A. undervalued. - Incorrect

Incorrect Answer Feedback: Incorrect because with an estimated value per share of $16.375, the shares are overvalued, not fairly valued or undervalued, because market price ($18.00) exceeds estimated value per share.

B. fairly valued. - Incorrect

Incorrect Answer Feedback: Incorrect because with an estimated value per share of $16.375, the shares are overvalued, not fairly valued or undervalued, because market price ($18.00) exceeds estimated value per share.

C. overvalued. - Correct

Correct Answer Feedback: Correct because estimated value (adjusted book value) per share = adjusted book value / number of shares outstanding = (market value of assets – market value of liabilities) / number of shares outstanding. Market value of assets = cash + accounts receivable + adjusted inventories + adjusted net fixed assets = 2.5 + 7.5+ (1.10 × 15.0) + (1.15 × 25) = 2.5 + 7.5 + 16.5 + 28.75 = 55.25. Adjusted book value = 55.25 – 22.5 = 32.75. Estimated value per share = 32.75 / 2.0 = $16.375. With a current share price of $18.00, the shares are overvalued because market price exceeds estimated value per share.

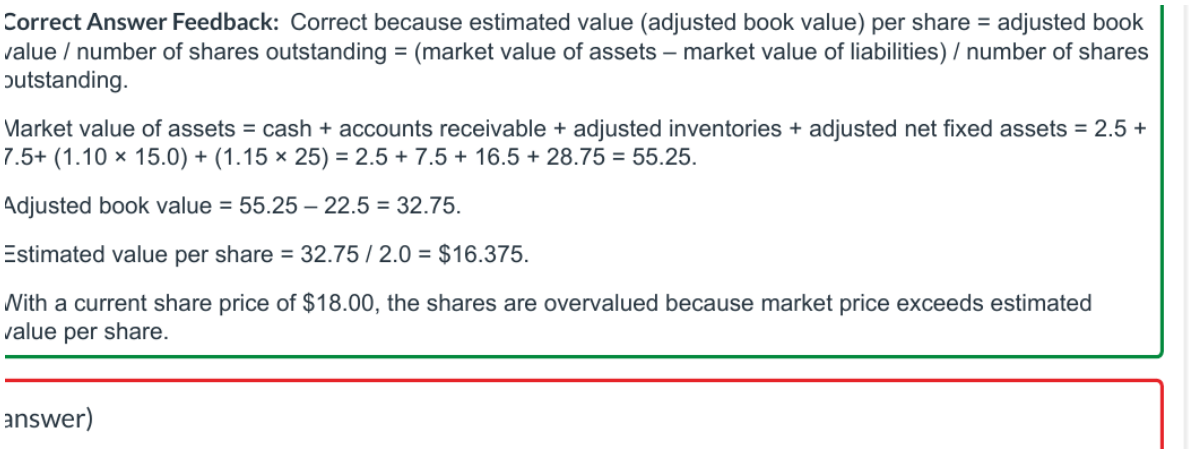


Figure 49: Source equation for question 66, option C

Question 67

Based on historical returns, a portfolio has a Sharpe ratio of 2.0. If the mean return to the portfolio is 20%, and the mean return to a risk-free asset is 4%, the standard deviation of return on the portfolio is closest to:

Correct answer: A

A. 8%. - Correct

Correct Answer Feedback: Correct. The Sharpe ratio for a portfolio p, based on historical returns, is defined as where is the mean return to the portfolio, is the mean return to a risk-free asset, and s is p the standard deviation of return on the portfolio. In this instance, 2 = (20% − 4%)/s . Solving for s : s = (20% − p p p 4%)/2 = 8%.

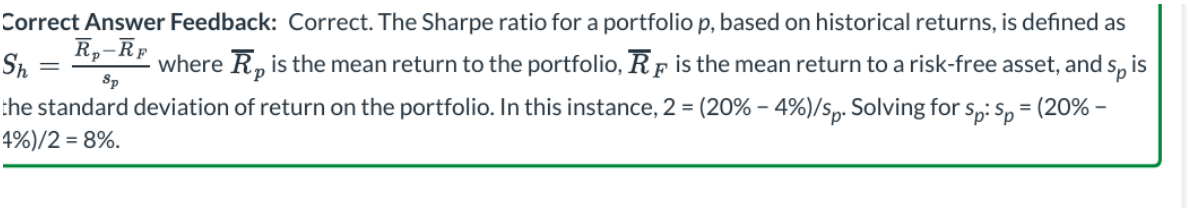


Figure 50: Source equation for question 67, option A

B. 10%. - Incorrect

Incorrect Answer Feedback: Incorrect. It fails to subtract the risk-free rate: (20%)/2 = 10%.

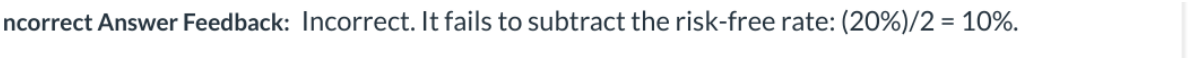


Figure 51: Source equation for question 67, option B

C. 12%. - Incorrect

Incorrect Answer Feedback: Incorrect. It adds the risk-free rate rather than subtracting it: (20% + 4%)/2 = 12%.

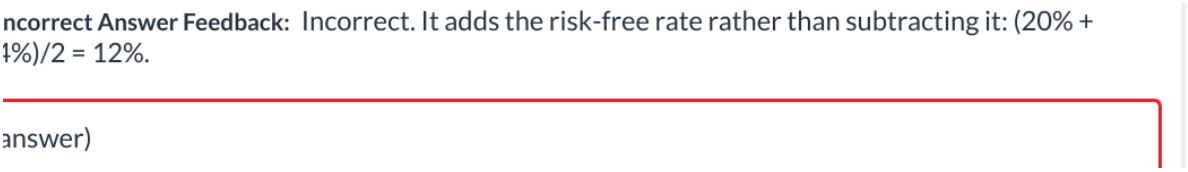


Figure 52: Source equation for question 67, option C

Question 68

With respect to return-generating models, statistical factor models:

Correct answer: B

A. only include factors that have economic meaning. - Incorrect

Incorrect Answer Feedback: Incorrect because in statistical factor models, the statistical factors, however, may or may not have an economic or fundamental connection to returns. Furthermore, in statistical factor models data mining may generate many spurious factors that are devoid of any economic meaning.

B. identify factors that explain the covariance in observed returns. - Correct

Correct Answer Feedback: Correct because in a statistical factor model, historical and cross-sectional return data are analyzed to identify factors that explain variance or covariance in observed returns.

C. only include factors that have a fundamental connection to returns. - Incorrect

Incorrect Answer Feedback: Incorrect because in statistical factor models, the statistical factors, however, may or may not have an economic or fundamental connection to returns. Furthermore, in statistical factor models data mining may generate many spurious factors that are devoid of any economic meaning.

Question 69

A portfolio consists of two securities with the following characteristics: Expected Return Standard Deviation of Returns Security 1 17% 24% Security 2 6% 12% Security 2 6% 12% If the portfolio has an expected return of 12.6% and the returns of the two securities are uncorrelated, the portfolio's standard deviation is closest to:

Correct answer: B

A. 13.4%. - Incorrect

Incorrect Answer Feedback: Incorrect because it uses an equally weighted portfolio of securities instead of deriving the securities' weights from the portfolio's expected return. Given that the two securities' returns are uncorrelated, it follows that the correlation between the two securities' returns is 0. Thus, portfolio standard deviation is calculated as: σ 2 = 0.52 × 0.242 + 0.52 × 0.122 p σp 2 = 0.0144 + 0.0036 σp 2 = 0.018 σ = √0.018 = 0.1342 ≈ 13.4%. p

ncorrect Answer Feedback: Incorrect because it uses an equally weighted portfolio of securities instead of leriving the securities' weights from the portfolio's expected return. Given that the two securities' returns are ıncorrelated, it follows that the correlation between the two securities' returns is 0. Thus, portfolio standard leviation is calculated as:

$$\sigma_p^2 = 0.5^2 \times 0.24^2 + 0.5^2 \times 0.12^2$$

$$\sigma_p^2 = 0.0144 + 0.0036$$

$$\sigma_p^2 = 0.018$$

$$\sigma_p = \sqrt{0.018} = 0.1342 \approx 13.4\%.$$

Figure 53: Source equation for question 69, option A

B. 15.2%. - Correct

Correct Answer Feedback: Correct because the portfolio's weights are calculated by setting the portfolio return equal to 12.6%. The portfolio return of a two-security portfolio is: R = w R + w R p 1 1 2 2 R = w R + (1 – w)R p 1 1 1 2 12.6% = w × 17% + (1 – w) × 6% 1 1 w = 0.6 and w = 1 – 0.6 = 0.4 1 2 The variance of a two-security portfolio is: σ 2 = (w 2σ 2 + w 2σ 2 + 2w w Cov(R , R))> p 1 1 2 2 1 2 1 2 Since the two securities' returns are uncorrelated, it follows that Cov(R , R) = 0, such that: 1 2 σp 2 = (0.62 × 0.242 + 0.42 × 0.122) σ 2 = (0.0207 + 0.0023) p σp = √0.0230 = 0.1518 ≈ 15.2%.

Correct Answer Feedback: Correct because the portfolio's weights are calculated by setting the portfolio return equal to 12.6%. The portfolio return of a two-security portfolio is:

$$R_p = w_1R_1 + w_2R_2$$

$$R_p = w_1R_1 + (1 - w_1)R_2$$

$$12.6\% = w_1 \times 17\% + (1 - w_1) \times 6\%$$

$w_1 = 0.6$ and $w_2 = 1 - 0.6 = 0.4$

The variance of a two-security portfolio is:

$$\sigma_p^2 = (w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2Cov(R_1, R_2))>$$

Since the two securities' returns are uncorrelated, it follows that $Cov(R_1, R_2) = 0$, such that:

$$\sigma_p^2 = (0.6^2 \times 0.24^2 + 0.4^2 \times 0.12^2)$$

$$\sigma_p^2 = (0.0207 + 0.0023)$$

$$\sigma_p = \sqrt{0.0230} = 0.1518 \approx 15.2\%.$$

Figure 54: Source equation for question 69, option B

C. 19.2%. - Incorrect

Incorrect Answer Feedback: Incorrect because it assumes that, since the securities' returns are uncorrelated, the portfolio standard deviation is the weighted mean of the individual securities' standard deviations. The portfolio's weights are calculated by setting the portfolio return equal to 12.6%. The portfolio return of a two- security portfolio is: R = w R + (1 – w)R p 1 1 1 2 12.6% = w × 17% + (1 – w) × 6% 1 1 w = 0.6 and w = 1 – 0.6 = 0.4 1 2 Then, σp is incorrectly calculated as: σ = 0.6 × 0.24 + 0.4 × 0.12 p σp = 0.192 = 19.2%. This would be the correct portfolio standard deviation if the correlation was equal to 1.

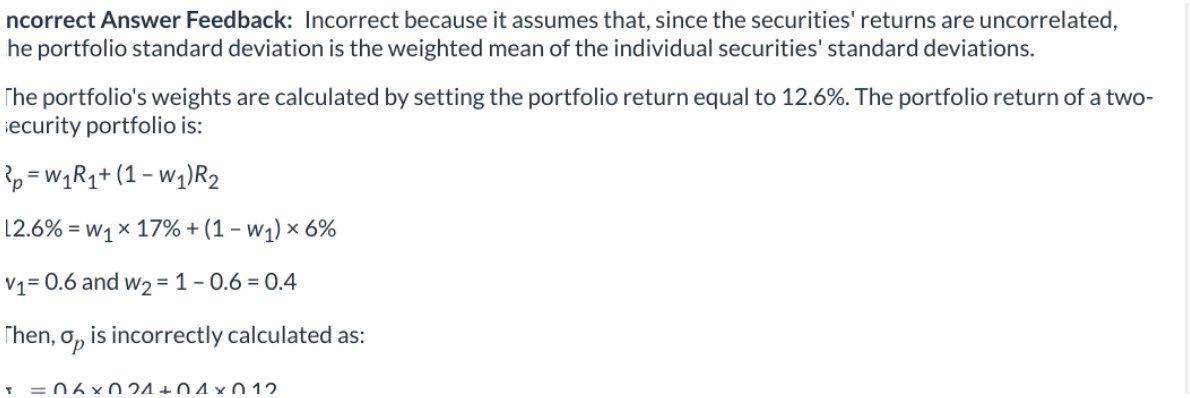


Figure 55: Source equation for question 69, option C

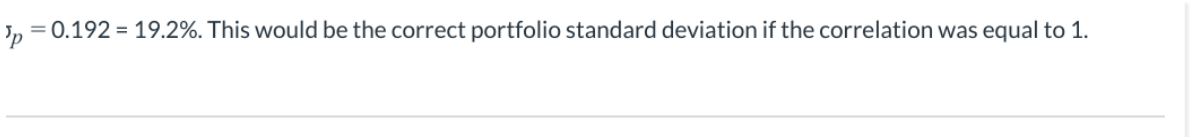


Figure 56: Source equation for question 69, option C

Question 70

A financial system is best described as operationally efficient when:

Correct answer: A

A. the costs of arranging trades are low. - Correct

Correct Answer Feedback: Correct because if the costs of arranging trades are low, the financial system is operationally efficient.

B. asset and contract prices reflect all available information. - Incorrect

Incorrect Answer Feedback: Incorrect because if the prices of the assets and contracts reflect all available information related to fundamental values, the financial system is informationally efficient.

C. an economy's resources are used where they are most valuable. - Incorrect

Incorrect Answer Feedback: Incorrect because economies that use resources where they are most valuable are allocationally efficient.

Question 71

If futures prices are positively correlated with interest rates then futures contracts are mostly likely:

Correct answer: B

A. less desirable to holders of long positions than are forwards. - Incorrect

Incorrect Answer Feedback: Incorrect because if futures prices are positively correlated with interest rates, futures contracts are more desirable, not less desirable, to holders of long positions than are forwards. The reason is because rising prices lead to futures profits that are reinvested in periods of rising interest rates, and falling prices leads to losses that occur in periods of falling interest rates. It is far better to receive cash flows in the interim than all at expiration under such conditions. This condition makes futures more attractive than forwards, and therefore their prices will be higher than forward prices.

B. more desirable to holders of long positions than are forwards. - Correct

Correct Answer Feedback: Correct because if futures prices are positively correlated with interest rates, futures contracts are more desirable to holders of long positions than are forwards. The reason is because rising prices lead to futures profits that are reinvested in periods of rising interest rates, and falling prices leads to losses that occur in periods of falling interest rates. It is far better to receive cash flows in the interim than all at expiration under such conditions. This condition makes futures more attractive than forwards, and therefore their prices will be higher than forward prices.

C. neither more or less desirable to holders of long positions than are forwards. - Incorrect

Incorrect Answer Feedback: Incorrect because if futures prices are positively correlated with interest rates, futures contracts are more desirable to holders of long positions than are forwards. The reason is because rising prices lead to futures profits that are reinvested in periods of rising interest rates, and falling prices leads to losses that occur in periods of falling interest rates. It is far better to receive cash flows in the interim than all at expiration under such conditions. This condition makes futures more attractive than forwards, and therefore their prices will be higher than forward prices.

Question 72

A security's market value and intrinsic value are most likely the same if:

Correct answer: A

A. the market is efficient. - Correct

Correct Answer Feedback: Correct because if investors believe a market is highly efficient, they will usually accept market prices as accurately reflecting intrinsic value.

B. active investment is profitable. - Incorrect

Incorrect Answer Feedback: Incorrect because if active investment is profitable, this suggests the market is inefficient, in which case, market value and intrinsic value are most likely different. If investors believe an asset market is relatively inefficient, they may try to develop an independent estimate of intrinsic value.

C. the security's cash flows are complex. - Incorrect

Incorrect Answer Feedback: Incorrect because the more complex an asset's future cash flows, the more difficult it is to estimate its intrinsic value.

Question 73

For a coupon bond with a positive yield, compounding more frequently within the year results in a yield-to- maturity that is:

Correct answer: A

A. less positive. - Correct

Correct Answer Feedback: Correct because for a given pair of cash flows, the stated annual rate and the periodicity are inversely related. Thus, increasing the frequency of compounding reduces the annual rate.

B. the same. - Incorrect

Incorrect Answer Feedback: Incorrect because for a given pair of cash flows, the stated annual rate and the periodicity are inversely related. Thus, increasing the frequency of compounding reduces the annual rate. The annual rate does not stay the same.

C. more positive. - Incorrect

Incorrect Answer Feedback: Incorrect because for a given pair of cash flows, the stated annual rate and the periodicity are inversely related. Thus, increasing the frequency of compounding reduces, not increases, the annual rate.

Question 74

The option-free bonds of Argus Corporation have a duration of eight years. When interest rates rise by 100 bps, the bond's price declines by 7.9%. When interest rates fall by 100 bps, however, the price rises by 8.2%. The asymmetrical price change is most likely caused by the:

Correct answer: C

A. coupon effect. - Incorrect

Incorrect Answer Feedback: Incorrect because the coupon effect relates to the sensitivity of bond price changes to changes in the coupon rate.

B. maturity effect. - Incorrect

Incorrect Answer Feedback: Incorrect because the maturity effect relates to the sensitivity of bond price changes to the time to maturity.

C. convexity effect. - Correct

Correct Answer Feedback: Correct because a fall in interest rates will result in a higher percentage rise in the bond's price compared with the percentage fall in the bond's price when interest rates rise by the same amount.

Question 75

In a binomial option pricing model, the expected payoff is based on:

Correct answer: B

A. estimated probabilities and discounted at the risk-free rate. - Incorrect

Incorrect Answer Feedback: Incorrect because, as described in the rationale for the correct answer, the binomial option pricing model uses risk-neutral probabilities.

B. risk-neutral probabilities and discounted at the risk-free rate. - Correct

Correct Answer Feedback: Correct because the value of an option today is its risk-neutral expected value at maturity, discounted at the risk-free rate. The risk-neutral expected value is determined using risk-neutral probabilities.

C. risk-neutral probabilities and discounted at the risk-free rate plus a risk premium. - Incorrect

Incorrect Answer Feedback: Incorrect because, as described in the rationale for the correct answer, the binomial option pricing model discounts the expected payoff at the risk-free rate.

Question 76

In the portfolio management process, the decision whether to use a nominal or real return objective is most likely part of the:

Correct answer: A

A. planning step. - Correct

Correct Answer Feedback: Correct because the planning step includes the preparation of an investment policy statement (IPS). One of the major components of an IPS is the return objective. Care should be taken that client and manager are in agreement on whether the return objective is nominal (which is more convenient for measurement purposes) or real (i.e., inflation-adjusted, which usually relates better to the objective).

B. feedback step. - Incorrect

Incorrect Answer Feedback: Incorrect because determination of the return objective is included in the preparation of the IPS, which is part of the planning, not feedback, step. The feedback step includes: (1) Portfolio monitoring and rebalancing, and (2) Performance measurement and reporting.

C. execution step. - Incorrect

Incorrect Answer Feedback: Incorrect because determination of the return objective is included in the preparation of the IPS, which is part of the planning, not execution, step. The execution step includes: (1) Asset allocation, (2) Security analysis, and (3) Portfolio construction.

Question 77

In relation to asset allocation, which of the following is a desirable property when specifying asset classes?

Correct answer: A

A. Asset classes should be mutually exclusive. - Correct

Correct Answer Feedback: Correct because when defining asset classes, a number of criteria apply. The asset classes, while being mutually exclusive, should add up to a sufficient approximation of the relevant investable universe.

B. An asset class should contain heterogeneous assets. - Incorrect

Incorrect Answer Feedback: Incorrect because asset classes should contain homogeneous assets, not heterogeneous assets. When defining asset classes, a number of criteria apply. Intuitively, an asset class should contain relatively homogeneous assets while providing diversification relative to other asset classes.

C. Each asset class should be composed of assets with low paired correlations. - Incorrect

Incorrect Answer Feedback: Incorrect because paired correlations of assets within an asset class should be high, not low. When defining asset classes, a number of criteria apply. In statistical terms, risk and return expectations should be similar and paired correlations of assets should be relatively high within an asset class but should be lower versus assets in other asset classes.

Question 78

With respect to a client's investment policy statement, the set of exposures to the permissible asset classes that is expected to achieve the client's long-term objectives best describes the:

Correct answer: C

A. security selection. - Incorrect

Incorrect Answer Feedback: Incorrect because security selection is an attempt to generate higher returns than the asset class benchmark by selecting securities with a higher expected return.

B. tactical asset allocation. - Incorrect

Incorrect Answer Feedback: Incorrect because the tactical asset allocation is the decision to deliberately deviate from the policy exposures to systematic risk factors (i.e., the policy weights of asset classes) with the intent to add value based on forecasts of the near-term returns of those asset classes.

C. strategic asset allocation. - Correct

Correct Answer Feedback: Correct because the strategic asset allocation (SAA) is the set of exposures to IPS-permissible asset classes that is expected to achieve the client's long-term objectives given the client's risk profile and investment constraints.

Question 79

When constructing an equity index, each company's weight in the index is dependent on its number of shares outstanding if the index is:

Correct answer: C

A. price weighted. - Incorrect

Incorrect Answer Feedback: Incorrect because in price weighting, the weight on each constituent security is determined by dividing its price by the sum of all the prices of the constituent securities and not by market cap, which is price multiplied by shares outstanding.

B. equal weighted. - Incorrect

Incorrect Answer Feedback: Incorrect because this method assigns an equal weight to each constituent security at inception and therefore the weight is not determined by market cap, which is price multiplied by shares outstanding.

C. market-capitilization weighted. - Correct

Correct Answer Feedback: Correct because in market-capitalization weighting, or value weighting, the weight on each constituent security is determined by dividing its market capitalization by the total market capitalization (the sum of the market capitalization) of all the securities in the index. Market capitalization or value is calculated by multiplying the number of shares outstanding by the market price per share.

Question 80

The methodology used by a credit rating agency to adjust a specific bond issue rating is best described as:

Correct answer: A

A. notching. - Correct

Correct Answer Feedback: Correct because specific issues may be assigned different credit ratings - higher or lower - due to a ratings adjustment methodology known as notching. For the rating agencies, likelihood of default —default risk—is the primary factor in assigning their ratings. However, there are secondary factors as well including priority of payment in the event of a default, potential loss severity in the event of default, structural subordination,

B. cross-default. - Incorrect

Incorrect Answer Feedback: Incorrect because although cross-default provisions, whereby events of default such as non-payment of interest on one bond trigger default on all outstanding debt, implies the same default probability for all issues, specific issues may be assigned different credit ratings - higher or lower - due to a ratings adjustment methodology known as notching. Notching is the process by which an issue credit rating is adjusted higher or lower, where as cross-default provisions describe a process whereby the default of a bond issue from the same issuer can trigger the default on other or all outstanding debt of that issuer. Cross-default provisions are terms in a bond issue and do not describe a credit rating methodology.

C. structural subordination. - Incorrect

Incorrect Answer Feedback: Incorrect because a factor considered by ratings agencies is structural subordination, which can arise when a corporation with a holding company structure has debt at both its parent holding company and operating subsidiaries. Debt at the operating subsidiaries will get serviced by the cash flow and assets of the subsidiaries before funds can be passed ("upstreamed") to the holding company to service debt at that level. Recognizing these different payment priorities, and thus the potential for higher (or lower) loss severity in the

event of default, the rating agencies have adopted a notching process whereby their credit ratings on issues can be moved up or down from the issuer rating. Notching is the process by which an issue credit rating is adjusted higher or lower, where as structural subordination is a factor that helps determine the notching outcome when a rating agency rates a bond issue.

Question 81

One source of return for an investor in a fixed-rate corporate bond is most likely the:

Correct answer: C

A. bid-ask spread. - Incorrect

Incorrect Answer Feedback: Incorrect because the bid–offer spread or bid–ask spread, which reflects the prices at which dealers will buy from a customer (bid) and sell to a customer (offer or ask), is very often used as an indicator of liquidity, not a source of return for an investor. The bid-ask spread may provide return for an intermediary such as a bank or broker selling the bond to the investor.

B. interest on the collateral. - Incorrect

Incorrect Answer Feedback: Incorrect because in contrast to corporate bonds, the source of payment for ABS does not depend on the claims-paying ability of an operating entity but on the cash flows generated by one or more underlying financial assets, such as mortgages or auto loans. Interest on the collateral of the bond may be a source of return from certain ABS, but not from a typical fixed-rate corporate bond.

C. reinvestment of coupon payments. - Correct

Correct Answer Feedback: Correct because the fixed-rate bond investor has three sources of return: (1) receipt of the promised coupon and principal payments on the scheduled dates, (2) reinvestment of coupon payments, and (3) potential capital gains or losses on the sale of the bond prior to maturity.

Question 82

The price of a fixed-rate corporate bond with an annual modified duration of 6.32 increases from 90 to 95 per 100 of par value. If the government benchmark yield declines by 5 bps, the estimated decline in the spread over the benchmark yield is closest to:

Correct answer: A

A. 82.9 bps. - Correct

Correct Answer Feedback: Correct because given that the price increases to 95 from 90, the percentage price increase is (95 – 90)/(90) ≈ 0.05556. Using %ΔPVFull ≈ –AnnModDur × ΔYield, the change in the yield-to- maturity can be calculated as 0.05556 = –6.32 × ΔYield (cid:15482) ΔYield = –0.05556/6.32 ≈ –0.00879, i.e. a decrease of 0.00879. Given that the yield-to-maturity on a corporate bond consists of a government benchmark yield and a spread, the decline in the government benchmark yield by 5 bps implies that the change in the spread over the benchmark decline in the government benchmark yield by 5 bps implies that the change in the spread over the benchmark yield is 0.00879 – 0.0005 = 0.00829, i.e. 82.9 bps, so that on a combined basis the decline in the yield-to-maturity is 0.0005 + 0.00829 = 0.00879.

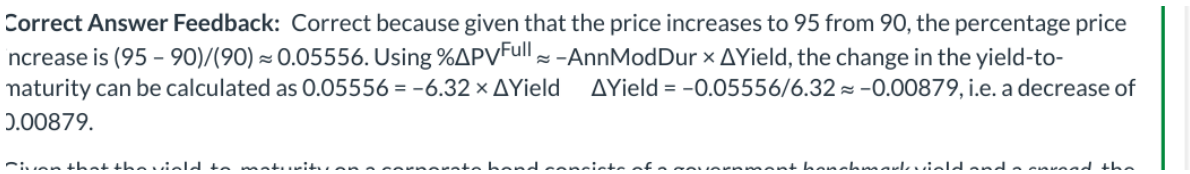

Correct Answer Feedback: Correct because given that the price increases to 95 from 90, the percentage price ncrease is (95 – 90)/(90) ≈ 0.05556. Using %ΔPV^{Full} ≈ –AnnModDur × ΔYield, the change in the yield-to- naturity can be calculated as 0.05556 = –6.32 × ΔYield ΔYield = –0.05556/6.32 ≈ –0.00879, i.e. a decrease of 0.00879.

Figure 57: Source equation for question 82, option A

decline in the government benchmark yield by 5 bps implies that the change in the spread over the benchmark yield is 0.00879 – 0.0005 = 0.00829, i.e. 82.9 bps, so that on a combined basis the decline in the yield-to-maturity is 0.0005 + 0.00829 = 0.00879.

Figure 58: Source equation for question 82, option A

B. 87.9 bps. - Incorrect

Incorrect Answer Feedback: Incorrect because it correctly calculates the decline in the yield-to-maturity of 0.00879 but omits to incorporate the effect of the government benchmark yield decrease, which means the decline in the spread over the benchmark yield is less than 87.9 bps, to have a total decrease in the yield-to- maturity of 87.9 bps.

C. 92.9 bps. - Incorrect

Incorrect Answer Feedback: Incorrect because it correctly calculates the decline in the yield-to-maturity of 0.00879 but instead of subtracting the decline in the benchmark yield to calculate the decline in the spread, it adds it back, i.e. 0.00879 + 0.0005 = 0.00929.

ncorrect Answer Feedback: Incorrect because it correctly calculates the decline in the yield-to-maturity of 0.00879 but instead of subtracting the decline in the benchmark yield to calculate the decline in the spread, it adds it back, i.e. 0.00879 + 0.0005 = 0.00929.

answer)

Figure 59: Source equation for question 82, option C

Question 83

Which of the following statements about REITs is most accurate?

Correct answer: C

A. Adding real estate to a portfolio of traditional investments will increase portfolio risk - Incorrect

Incorrect Answer Feedback: Incorrect because real estate historically has low correlations with other asset classes, and adding real estate to an investment portfolio provides diversification benefits at relatively lower levels of risk.

B. Adding real estate to a portfolio of traditional investments will decrease portfolio diversification - Incorrect

Incorrect Answer Feedback: Incorrect because real estate historically has low correlations with other asset classes, and adding real estate to an investment portfolio provides diversification benefits at relatively lower levels of risk.

C. Equity REIT correlations with traditional investments are at their highest during steep market downturns - Correct

Correct Answer Feedback: Correct because during certain market conditions, equity REIT correlations with market benchmarks increase, particularly during steep market downturns, such as the 2007–08 financial crisis. As these high correlations remained high during the post-crisis recovery, they lifted the value of most asset classes. Similar patterns in financial asset prices and real estate prices appear to have been observed during the COVID-19 recovery.

Question 84

An analyst gathers the following data for a 2-year option contract: Forward price 1.2000 USD/EUR Strike price 1.2250 USD/EUR Risk-free rate 5% Call premium 0.0500 USD/EUR Call premium 0.0500 USD/EUR USD/EUR is the amount of USD per 1 EUR The put premium is closest to:

Correct answer: B

A. 0.0273 USD/EUR. - Incorrect

Incorrect Answer Feedback: Incorrect because this answer mistakenly flips the strike price and the forward price in the equation p = c + [X – F (T)] / (1 + r)T. This incorrect answer is calculated as p = 0.0500 + (1.2000 – 0 0 0 0 1.2250) / (1 + 0.05)2 = 0.0273.

ncorrect Answer Feedback: Incorrect because this answer mistakenly flips the strike price and the forward ›rice in the equation $p_0 = c_0 + [X - F_0(T)] / (1 + r)^T$. This incorrect answer is calculated as $p_0 = 0.0500 + (1.2000 - 1.2250) / (1 + 0.05)^2 = 0.0273$.

Figure 60: Source equation for question 84, option A

B. 0.0727 USD/EUR. - Correct

Correct Answer Feedback: Correct because put–call forward parity can be expressed as: F (T)(1 + r)-T + p = c + X(1 + r)-T, 0 0 0 where F (T)(1 + r)-T is the present value of F (T) discounted at the risk-free rate, p is the price of the put option 0 0 0 on the underlying at t = 0, c is the price of the call option on the underlying at t = 0, X(1 + r)-T is a risk-free bond 0 that pays the amount of the exercise price X at t = T. In other words, under put–call parity, at t = 0 the price of the long underlying asset plus the long put must equal the price of the long call plus the risk-free asset. Rearranging the formula yields: F (T)(1 + r)-T – X(1 + r)-T = c – p 0 0 0 [F (T) – X] / (1 + r)T = c – p 0 0 0 p = c + [X – F (T)] / (1 + r)T 0 0 0 Substituting in yields: p = 0.0500 + (1.2250 – 1.2000) / (1 + 0.05)2 = 0.0727. 0

Correct Answer Feedback: Correct because put–call forward parity can be expressed as:

$F_0(T)(1 + r)^{-T} + p_0 = c_0 + X(1 + r)^{-T}$,

where $F_0(T)(1 + r)^{-T}$ is the present value of $F_0(T)$ discounted at the risk-free rate, p_0 is the price of the put option on the underlying at $t = 0$, c_0 is the price of the call option on the underlying at $t = 0$, $X(1 + r)^{-T}$ is a risk-free bond that pays the amount of the exercise price X at $t = T$.

In other words, under put–call parity, at $t = 0$ the price of the long underlying asset plus the long put must equal the price of the long call plus the risk-free asset.

Rearranging the formula yields:

$F_0(T)(1 + r)^{-T} - X(1 + r)^{-T} = c_0 - p_0$

$[F_0(T) - X] / (1 + r)^T = c_0 - p_0$

$p_0 = c_0 + [X - F_0(T)] / (1 + r)^T$

Substituting in yields:

$p_0 = 0.0500 + (1.2250 - 1.2000) / (1 + 0.05)^2 = 0.0727$.

Figure 61: Source equation for question 84, option B

C. 0.0750 USD/EUR. - Incorrect

Incorrect Answer Feedback: Incorrect because this fails to take the present value of the strike price minus the forward price. In other words, it fails to divide by (1 + r)T in the equation p = c + [X – F (T)] / (1 + r)T. This 0 0 0 incorrect answer is calculated as p = 0.0500 + (1.2250 – 1.2000) = 0.0750. 0

ncorrect Answer Feedback: Incorrect because this fails to take the present value of the strike price minus the orward price. In other words, it fails to divide by $(1 + r)^T$ in the equation $p_0 = c_0 + [X - F_0(T)] / (1 + r)^T$. This ncorrect answer is calculated as $p_0 = 0.0500 + (1.2250 - 1.2000) = 0.0750$.

ıswer)

Figure 62: Source equation for question 84, option C

Question 85

For non-amortizing, non-mortgage asset-backed securities, the lockout period most likely represents when:

Correct answer: C

A. investors start receiving repayments. - Incorrect

Incorrect Answer Feedback: Incorrect. When the lockout period is over, any principal that is repaid will not be used to reinvest in new loans but will instead be distributed to the bond classes.

B. overcollateralization begins to be reduced. - Incorrect

Incorrect Answer Feedback: Incorrect. Overcollateralization refers to a reserve account, often an excess spread account.

C. repaid principal is reinvested in loans of equal principal. - Correct

Correct Answer Feedback: Correct. The lockout period, or revolving period, is the period during which the principal repaid is reinvested to acquire additional loans with a principal equal to the principal repaid. When the lockout period is over, any principal that is repaid will not be used to reinvest in new loans but will instead be distributed to the bond classes. Overcollateralization refers to a reserve account, often an excess spread account.

Question 86

An analyst gathers the following information about a bond portfolio:

	Par Value	Book Value	Market Value	Modified Duration
Bond 1	$700,000	$750,000	$770,000	3
Bond 2	$825,000	$700,000	$850,000	8

The portfolio's duration is closest to:

Correct answer: B

A. 5.4. - Incorrect

Incorrect Answer Feedback: Incorrect because book value was used instead of market value. The Macaulay and modified durations for the portfolio are calculated as the weighted average of the statistics for the individual bonds. The shares of overall portfolio market value are the weights. ≠ (750,000/1,450,000) x 3 + (700,000/1,450,000) x 8 = 1.5517 + 3.8621 = 5.4138 ≈ 5.4.

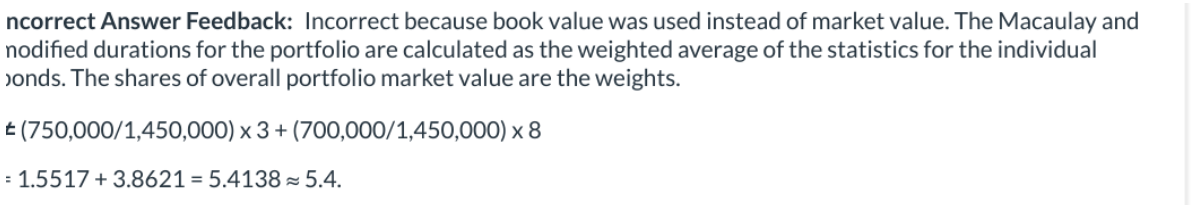


Figure 63: Source equation for question 86, option A

B. 5.6. - Correct

Correct Answer Feedback: Correct because the Macaulay and modified durations for the portfolio are calculated as the weighted average of the statistics for the individual bonds. The shares of overall portfolio market value are the weights. Portfolio Dur = (MV /PMV)xDur + (MV /PMV)xDur 1 1 2 2 = (770,000/1,620,000) x 3 + (850,000/1,620,000) x 8 = 1.4259 + 4.1975 = 5.6234 ≈ 5.6.

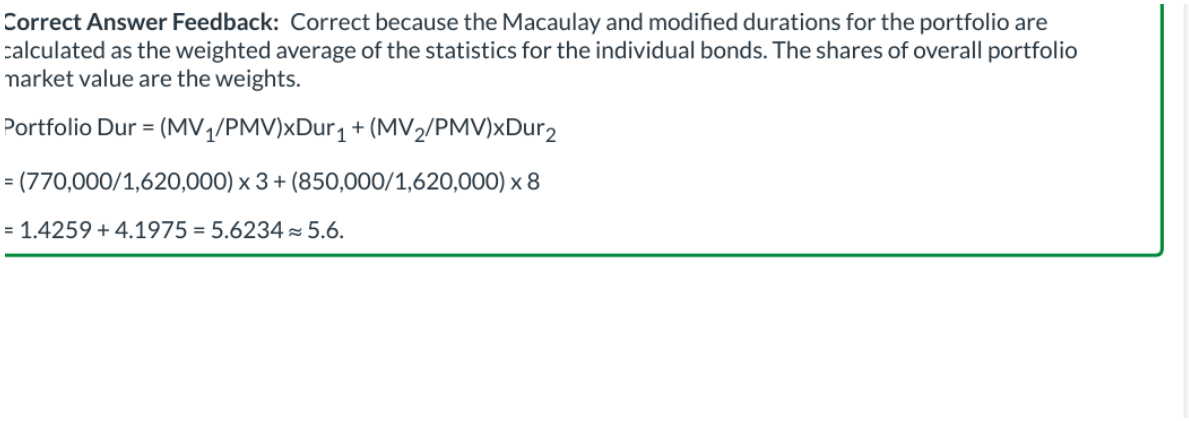


Figure 64: Source equation for question 86, option B

C. 5.7. - Incorrect

Incorrect Answer Feedback: Incorrect because par value was used instead of market value. The Macaulay and modified durations for the portfolio are calculated as the weighted average of the statistics for the individual bonds. The shares of overall portfolio market value are the weights. ≠ (700,000/1,525,000) x 3 + (825,000/1,525,000) x 8 = 1.3770 + 4.3279 = 5.7049 ≈ 5.7.

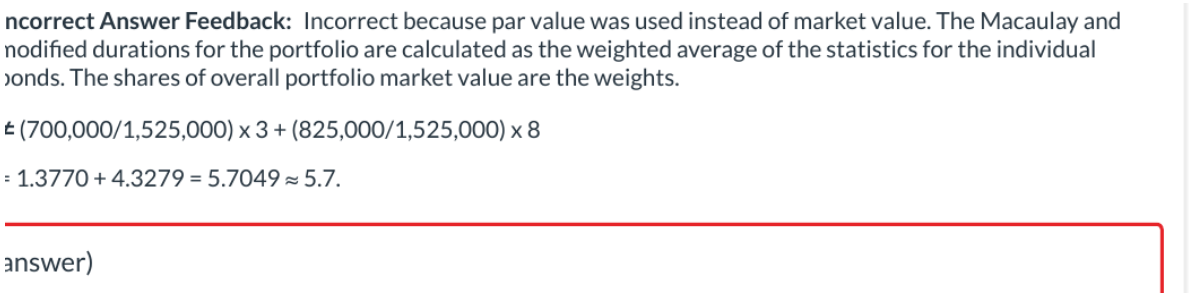


Figure 65: Source equation for question 86, option C

Question 87

An investor purchases common shares at €22 per share on margin and deposits just enough money to cover the 60% initial margin requirement. If the maintenance margin requirement is 25%, the share price below which the investor will first receive a margin call is closest to: investor will first receive a margin call is closest to:

Correct answer: A

A. €11.73. - Correct

Correct Answer Feedback: Correct because the investor deposits 60% of the purchase price (€13.20) as initial margin and borrows the balance (€8.80) from the brokerage company. The share price below which the investor will receive a margin call is computed as follows: Price = borrowing/share divided by (1 – maintenance margin requirement) €11.73 = €8.80 / (1 – 0.25).



Figure 66: Source equation for question 87, option A

B. €16.50. - Incorrect

Incorrect Answer Feedback: Incorrect because the purchase price is multiplied by (1 minus the maintenance margin requirement). €16.50 = €22 × (1 – 0.25).

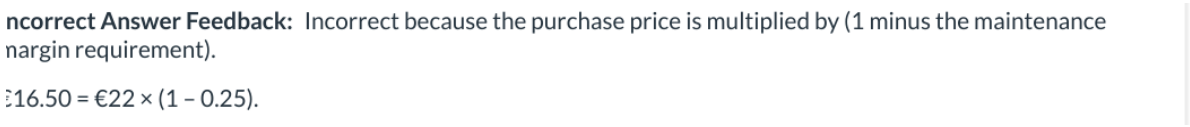


Figure 67: Source equation for question 87, option B

C. €17.60. - Incorrect

Incorrect Answer Feedback: Incorrect because the amount borrowed is computed as 60% of the purchase price (€13.20). €17.60 = €13.20 / (1 – 0.25).

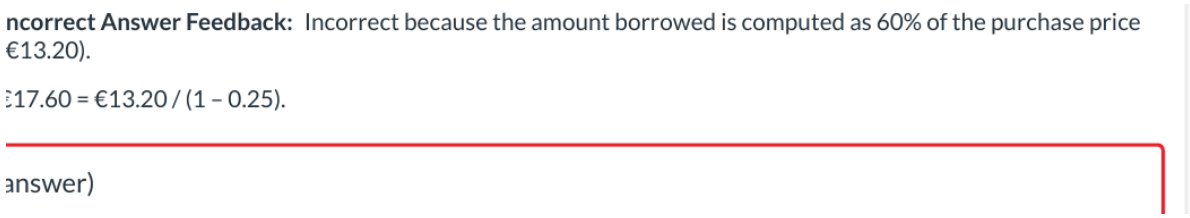


Figure 68: Source equation for question 87, option C

Question 88

An analyst gathers the following information about the forward curve for one-year rates: Time Period Forward Rate 0y1y 1.20% 1y1y 1.88% 2y1y 2.46% 3y1y 2.93% The price of a three-year, 5% annual-pay bond is closest to:

Correct answer: B

A. 107.42. - Incorrect

Incorrect Answer Feedback: Incorrect because the bond also can be valued using the forward curve. This answer incorrectly begins with the 1y1y forward rate, rather than the 0y1y rate. 5 / 1.0188 + 5 / (1.0188 × 1.0246) + 105 / (1.0188× 1.01246 × 1.0293) = 4.91 + 4.79 + 97.72 = 107.42 ≈ 107.4.

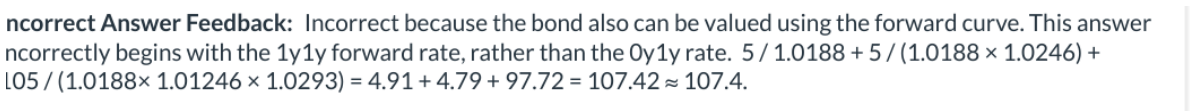


Figure 69: Source equation for question 88, option A

B. 109.18. - Correct

Correct Answer Feedback: Correct because the bond also can be valued using the forward curve. 5 / 1.012 + 5 / (1.012 × 1.0188) + 105 / (1.012 × 1.0188 × 1.0246) = 4.94 + 4.85 + 99.39 = 109.18 ≈ 109.2.

C. 111.73. - Incorrect

Incorrect Answer Feedback: Incorrect because the bond also can be valued using the forward curve. This answer adds the total payments over the three years and divides by 3y1y. 115 / 1.02932.93 = 111.73.

Question 89

An American waterfall distributes performance fees on a(n):

Correct answer: A



Figure 70: Source equation for question 88, option B

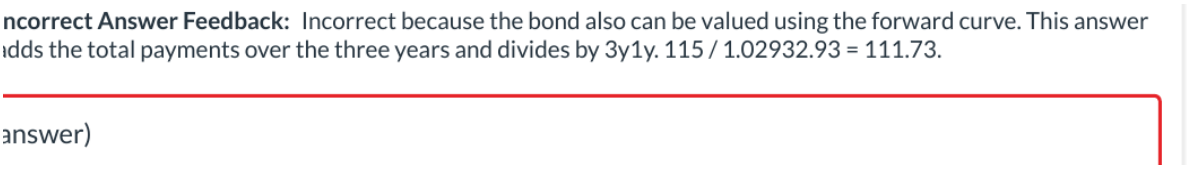


Figure 71: Source equation for question 88, option C

A. deal-by-deal basis and is more advantageous to the general partner than a European waterfall. - Correct

Correct Answer Feedback: Correct because there are two types of waterfalls: deal-by-deal (or American) waterfalls and whole-of-fund (or European) waterfalls. Deal-by-deal waterfalls are more advantageous to the GP because performance fees are collected on a per-deal basis, allowing the GP to get paid before LPs receive both their initial investment and their preferred rate of return (i.e., the hurdle rate) on the entire fund.

B. deal-by-deal basis and is more advantageous to the limited partners than a European waterfall. - Incorrect

Incorrect Answer Feedback: Incorrect because there are two types of waterfalls: deal-by-deal (or American) waterfalls and whole-of-fund (or European) waterfalls. In contrast to deal-by-deal waterfalls, whole-of-fund waterfalls occur at the aggregate fund level (i.e., after all investments have been exited) and are more advantageous to the LPs.

C. aggregated fund level and is more advantageous to the limited partners than a European waterfall. - Incorrect

Incorrect Answer Feedback: Incorrect because there are two types of waterfalls: deal-by-deal (or American) waterfalls and whole-of-fund (or European) waterfalls. In contrast to deal-by-deal waterfalls, whole-of-fund waterfalls occur at the aggregate fund level (i.e., after all investments have been exited) and are more advantageous to the LPs.

Question 90

A factor that most likely measures a client's ability to bear risk is his or her:

Correct answer: A

A. time horizon. - Correct

Correct Answer Feedback: Correct. A longer time horizon tends to imply greater ability to bear risk.

B. personality type. - Incorrect

Incorrect Answer Feedback: Incorrect. Personality type gives an indication about the willingness (not ability) of a client to bear risk.

C. inclination to independent thinking. - Incorrect

Incorrect Answer Feedback: Incorrect. Inclination to independent thinking gives an indication about the willingness (not ability) of a client to bear risk.